

iNKT: cell states, cytotoxic programs and functional changes

Integrated editable presentation | main talk + complete source-page appendix

Research question

How does the T2-associated response differ across tissues and iNKT states?

Presentation sequence

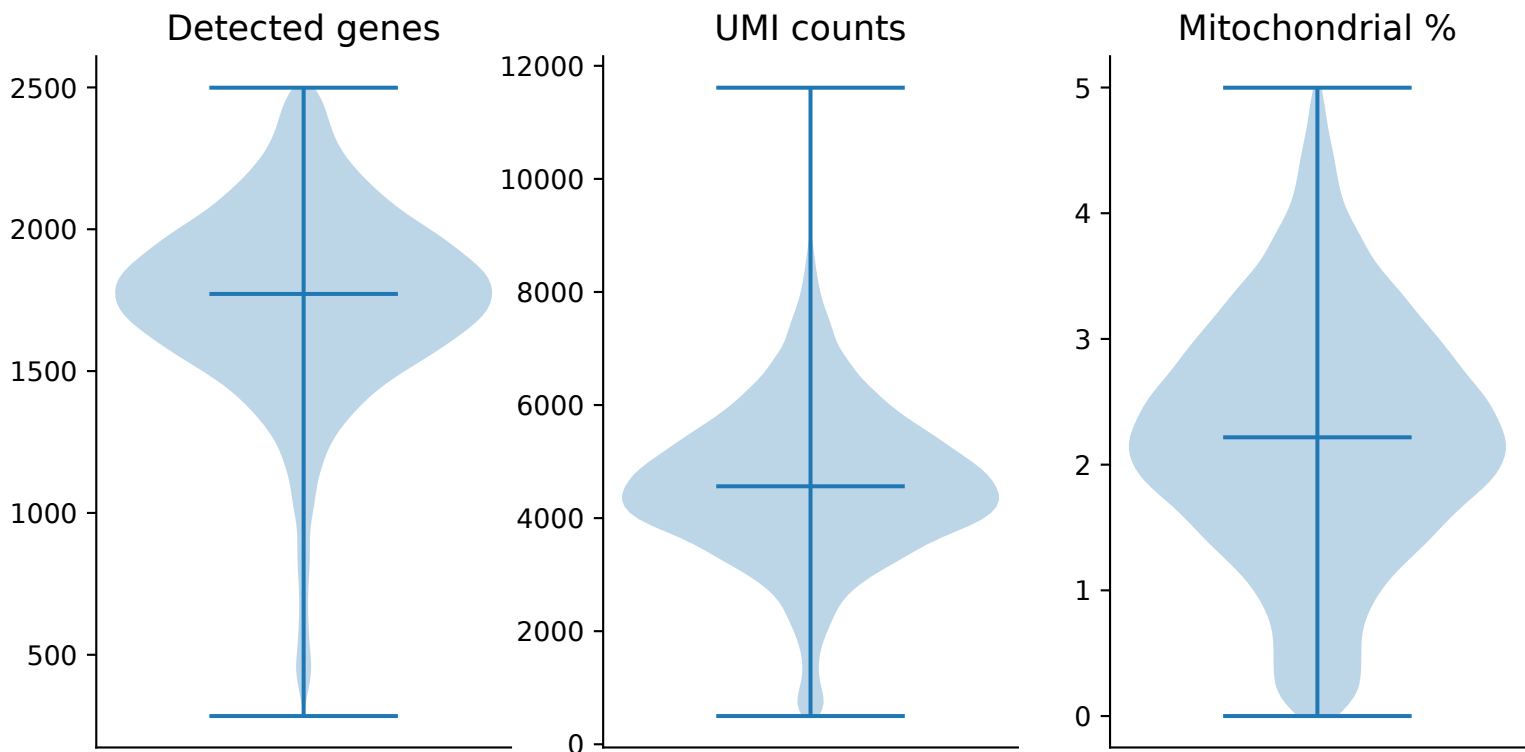
- 1 Cohort, cell-state markers and tissue composition
- 2 Differential expression, cytotoxic modules and GO functions
- 3 Local state ordering, robustness and bounded interpretation

15,532 cells | 3 tissues | Ctrl / T2 | 9 original refined clusters
One sample label per tissue x condition: all biological conclusions remain exploratory.

Material, dataset and QC

Six samples • 15,532 retained cells • 10,670 measured genes

Sample	Input cells	Retained
Ctrl_Thymus	2101	1372
Ctrl_BM	3998	3379
Ctrl_Spleen	3829	3422
T2_Thymus	1639	1257
T2_BM	3131	2731
T2_Spleen	3760	3371

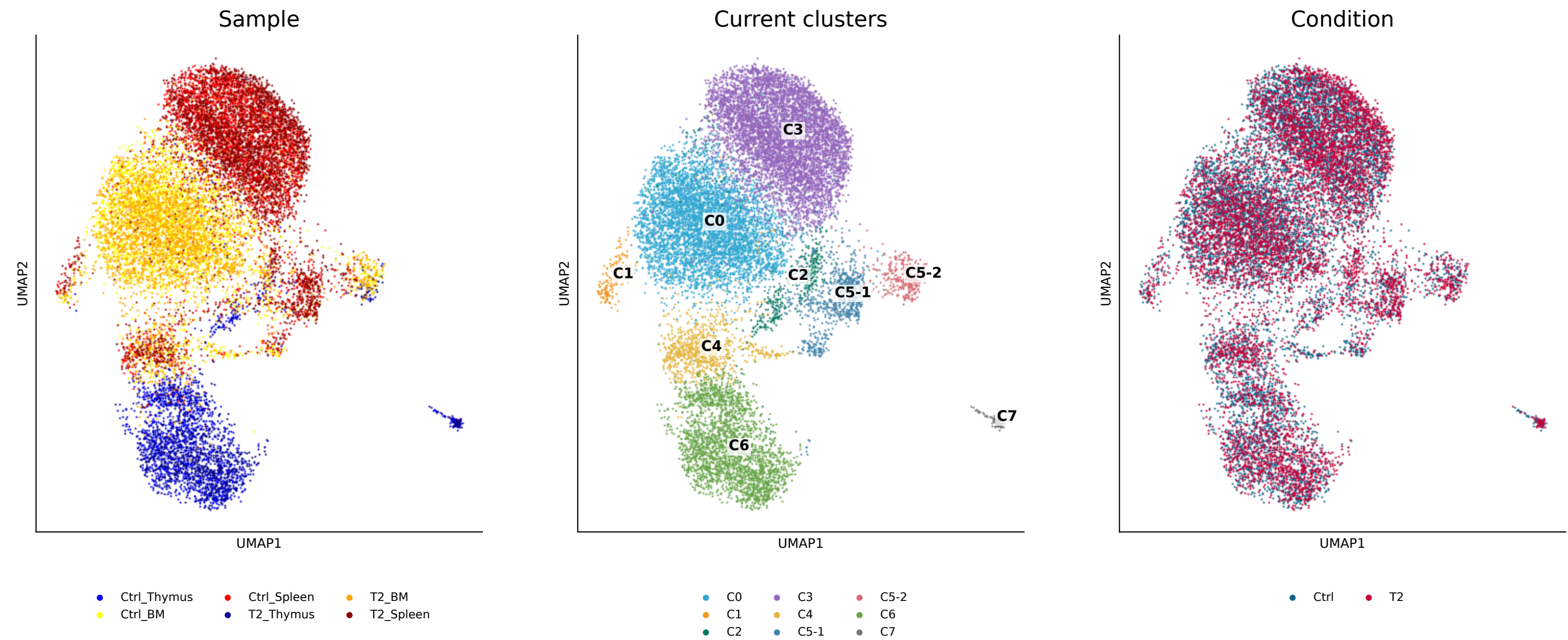


Gene min_cells = 100, then
 $200 \leq \text{detected genes} < 2,500$; mitochondrial fraction $< 5\%$.

QC counts match the legacy deck; downstream UMAP, Leiden and gene statistics remain a reanalysis.

Clusters and tissue composition

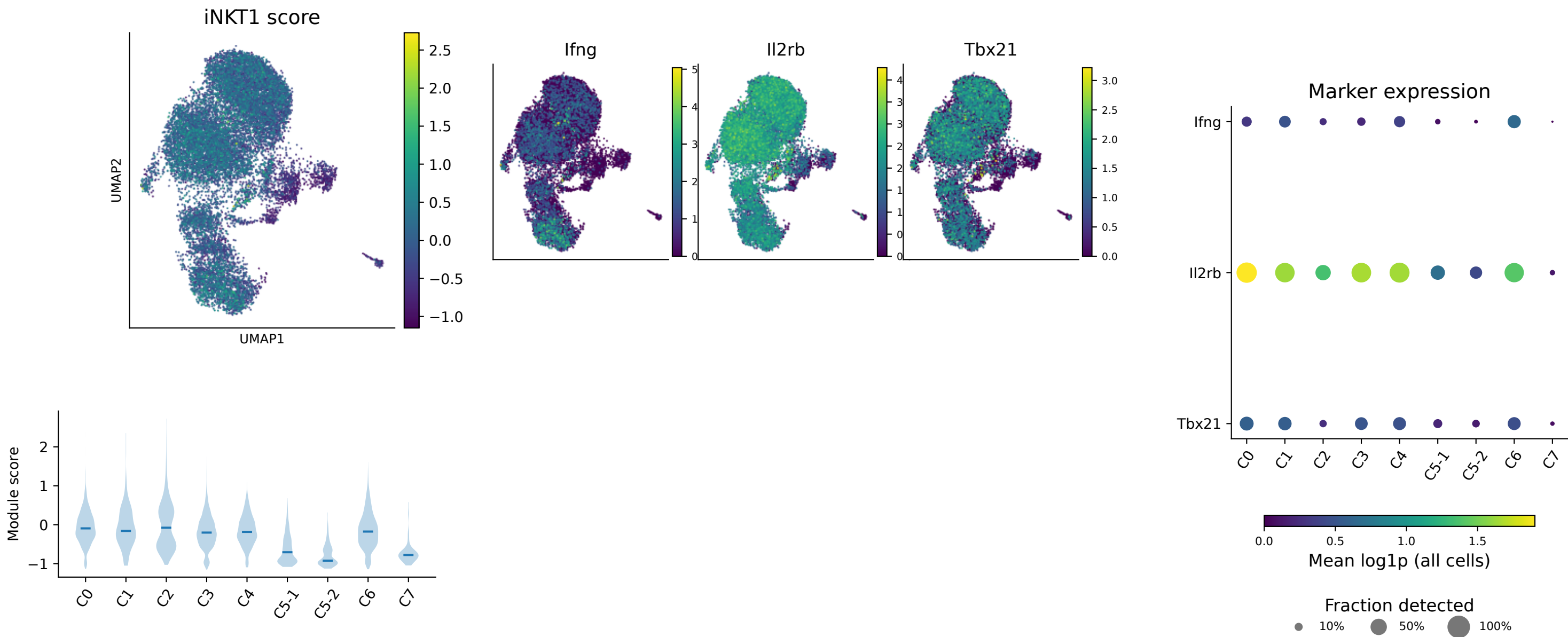
Fixed 50-PC UMAP; C5 refined to C5-1 and C5-2



Original sample hues retained on the left; Blood-style blue/red encodes condition on the right. Coordinates are unchanged.

iNKT1 marker program

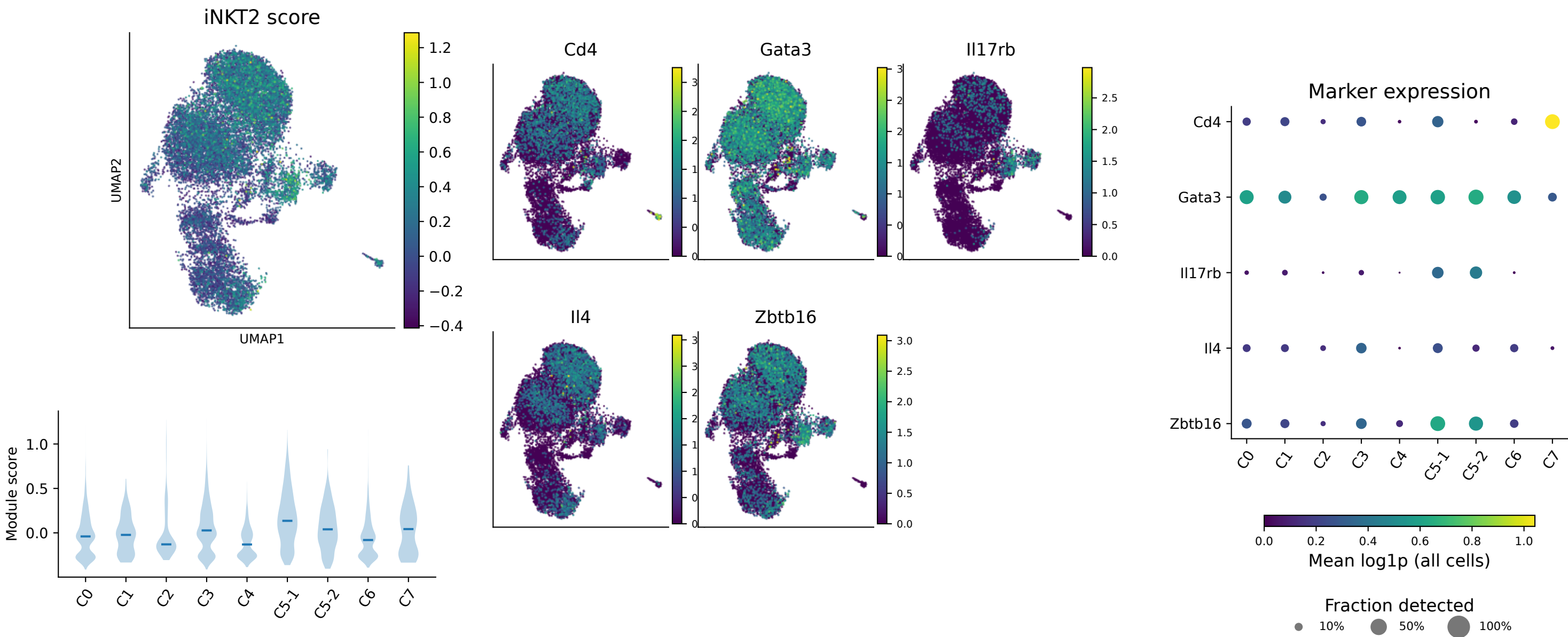
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

iNKT2 marker program

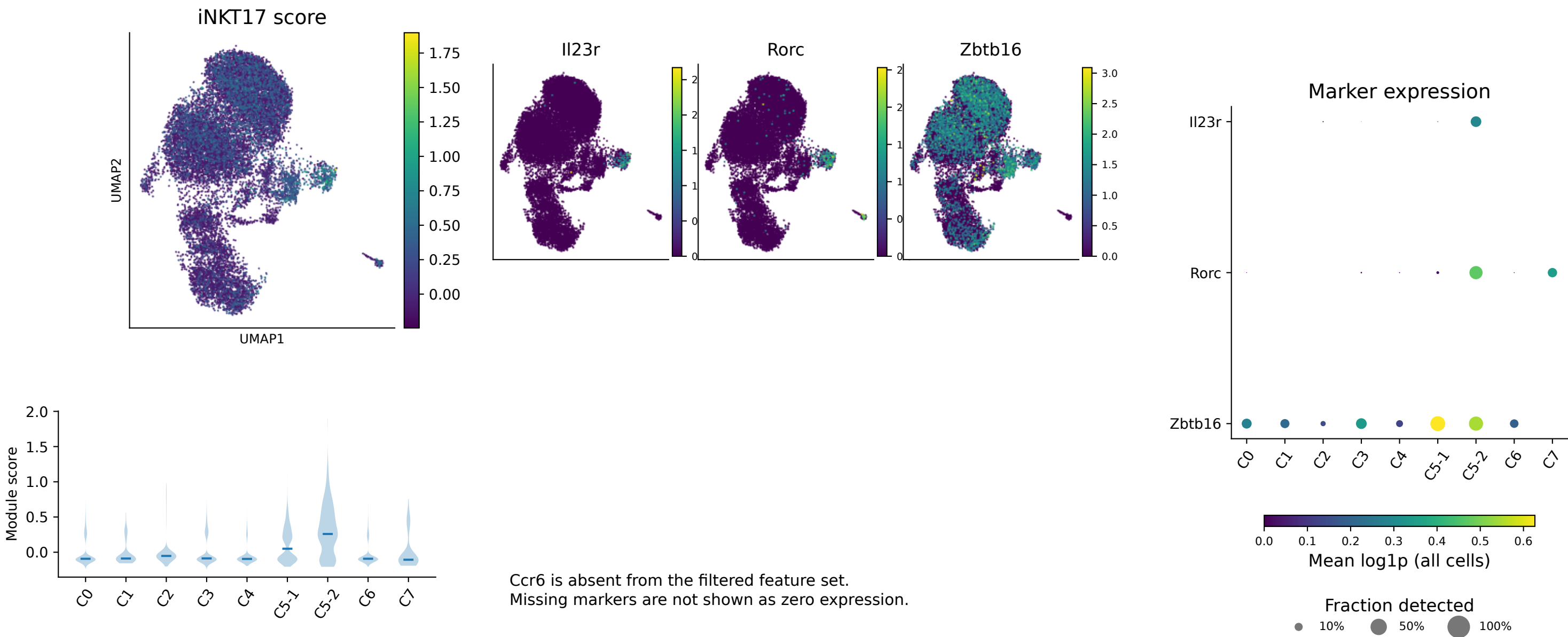
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

iNKT17 marker program

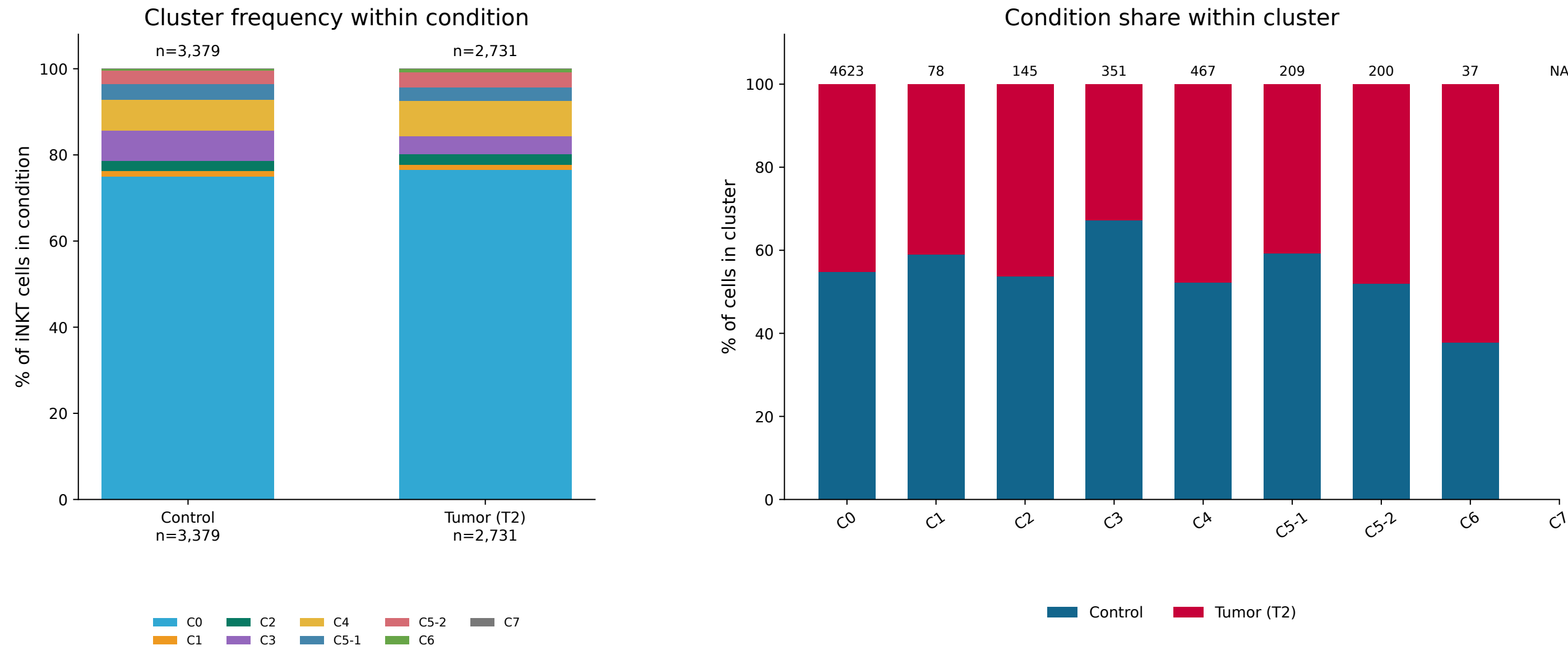
Score map, cluster distributions, individual markers and expression dotplot



Current score_genes units differ from legacy summed-marker scores; these panels do not establish discrete subtype identities.

Bone marrow: cluster and condition frequencies

C • The two denominators from Blood Figure 3C, presented side by side

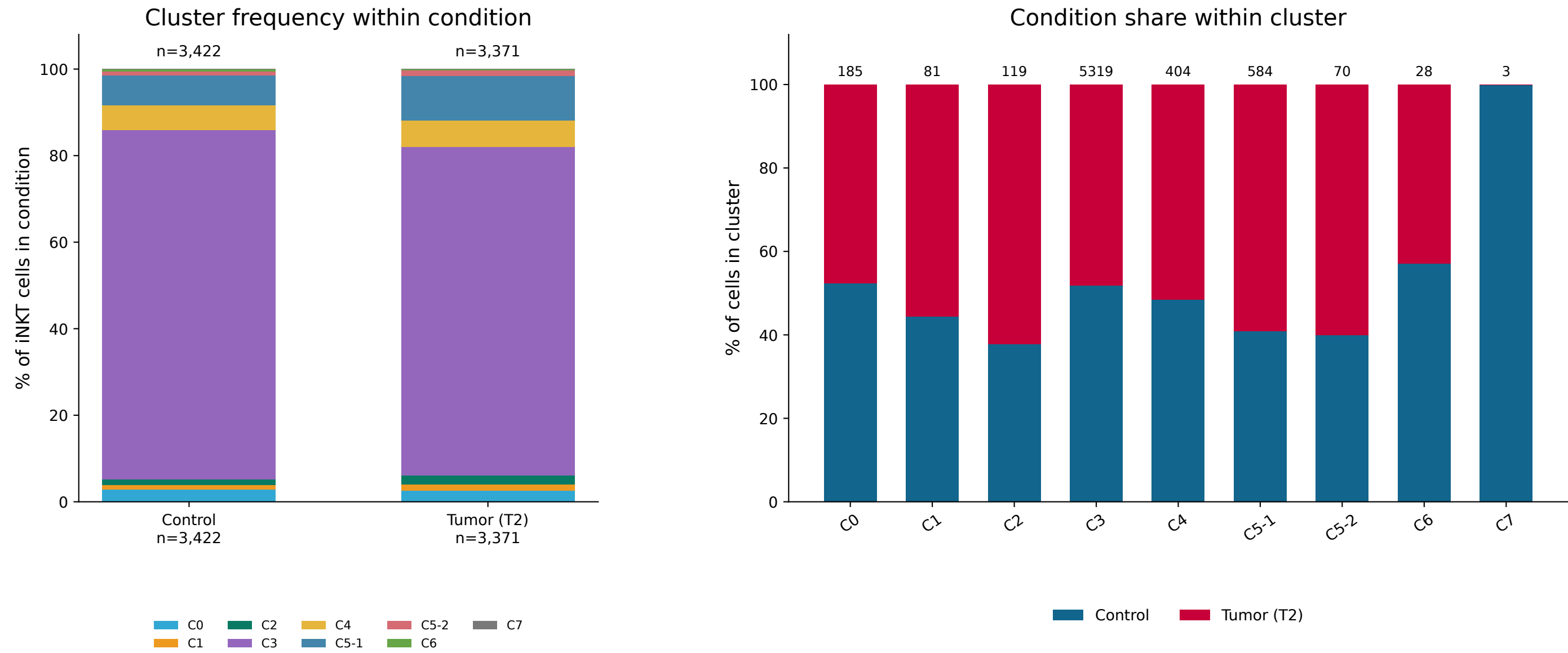


Left: $n(\text{cluster, condition}) / N(\text{condition})$ | Right: $n(\text{condition, cluster}) / N(\text{cluster})$

Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

Spleen: cluster and condition frequencies

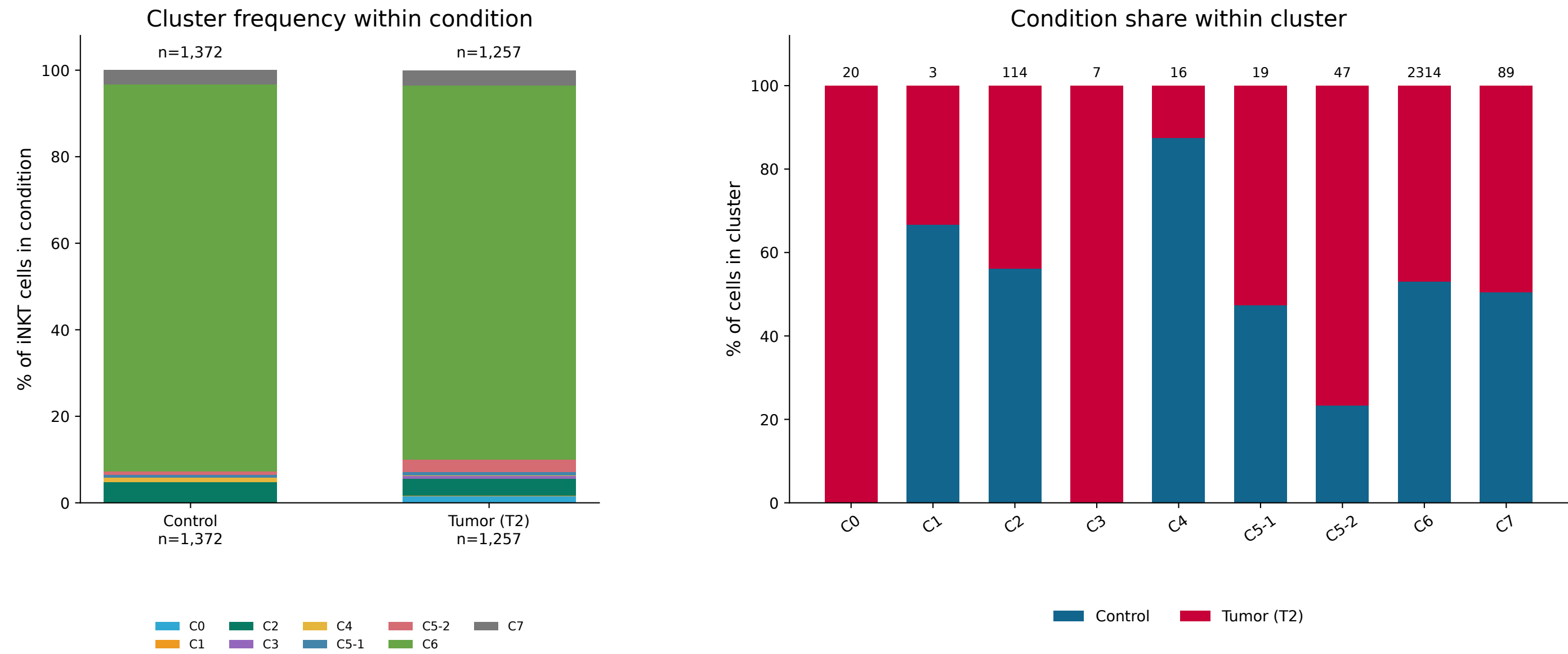
C • The two denominators from Blood Figure 3C, presented side by side



Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

Thymus: cluster and condition frequencies

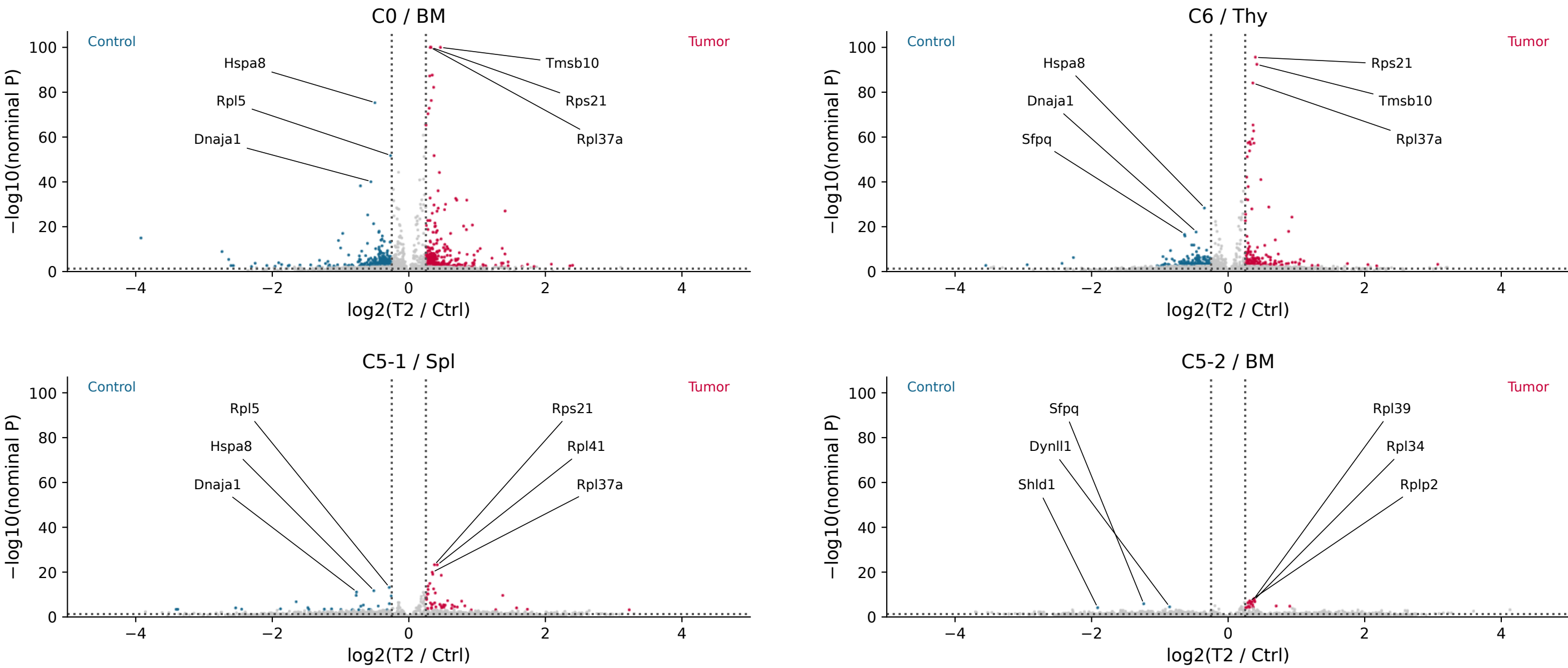
C • The two denominators from Blood Figure 3C, presented side by side



Captured-cell proportions only. No mouse-level error bars or significance are inferred from individual cells.

D Differential expression: Control vs. Tumor

Paper axes and blue/red direction; current tissue × cluster comparisons



Common axes: $|\log_2\text{FC}| \leq 5$; $-\log_{10}(P)$ capped at 100. All uncapped values and display flags are exported.

Color rule retained: gene BH FDR ≤ 0.05 and $|\log_2\text{FC}| \geq 0.25$. Vertical display thresholds are therefore ± 0.25 .

Analysis design and reference mapping

Methods adapted to mouse iNKT; biological conclusions are derived from these data.

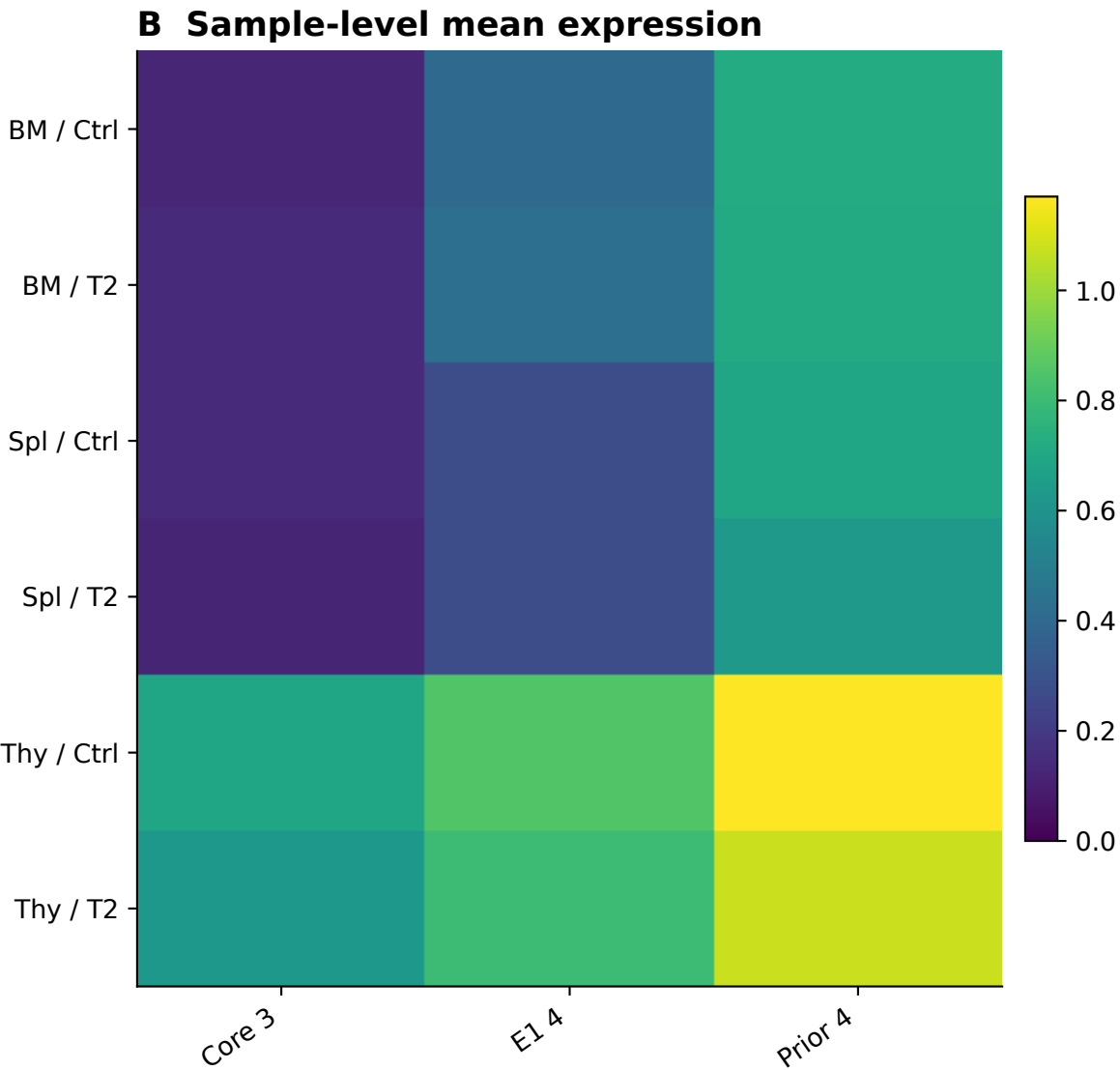
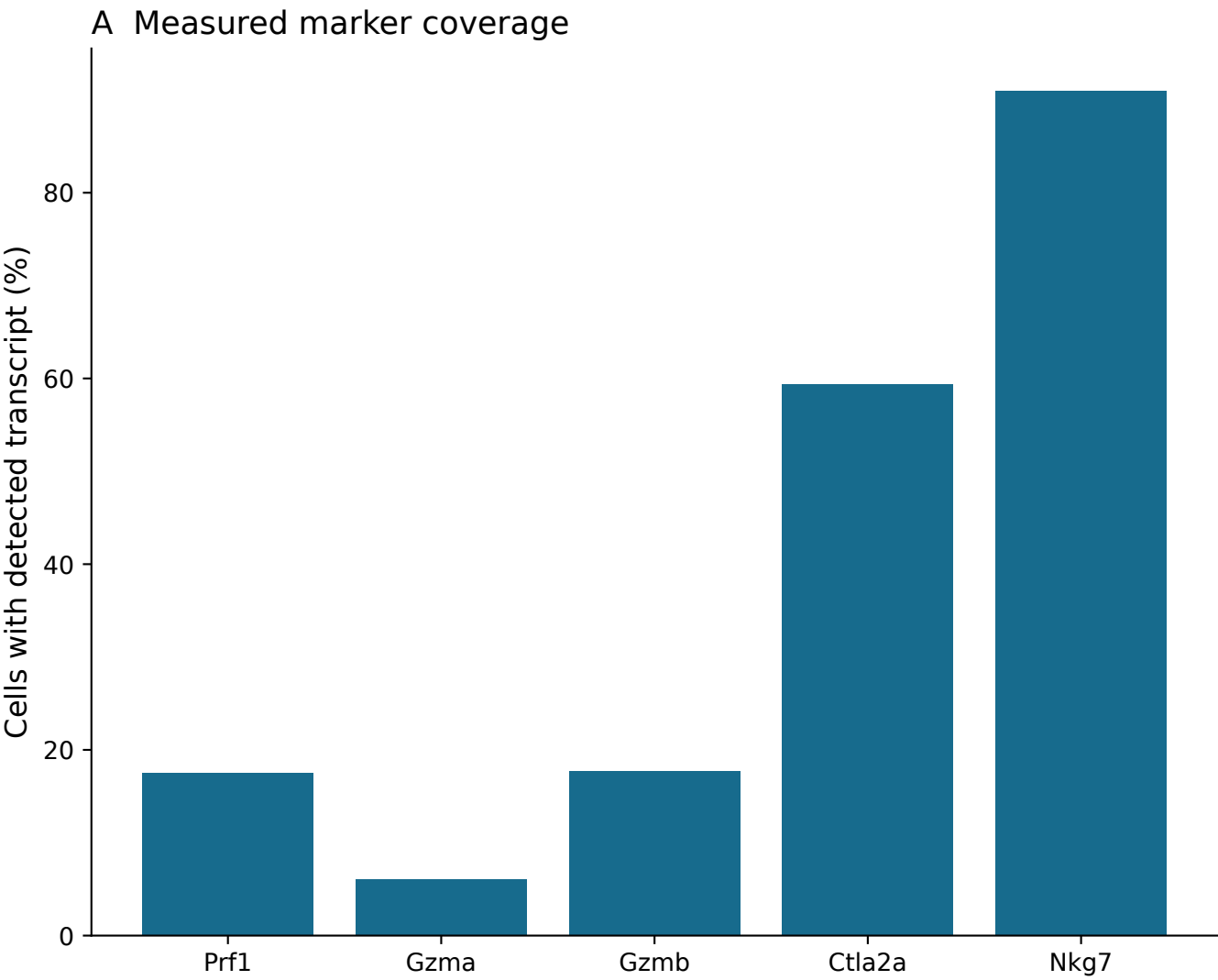
Analysis	Reference	Input / definition	Implementation
Cytotoxicity	PAGER Fig. 5C/D	Core: Prf1, Gzma, Gzmb; E1: +Ctla2a	Mean log1p expression; matched-control score sensitivity
GO cluster identity	GAFA Fig. 4c	Cluster vs rest within each tissue	Wilcoxon, gene BH <= .05 and log2FC >= .25
GO treatment response	GAFA functional comparison	T2-up and Ctrl-up genes tested separately	Hypergeometric; full eligible GO BP/MF/CC families
Local trajectories	PAGER Fig. 5D	Recompute diffusion map on tissue-specific graphs	Precursor-rooted DPT; 20 shared bins; 3-root sensitivity
Robustness	Adaptation to this dataset	Stable clusters; five QC-matched resamples	Effects and direction consistency; no animal replication

GO uses official mouse annotations. An enrichment of upregulated genes does not prove pathway activation.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Cytotoxic gene coverage and scoring

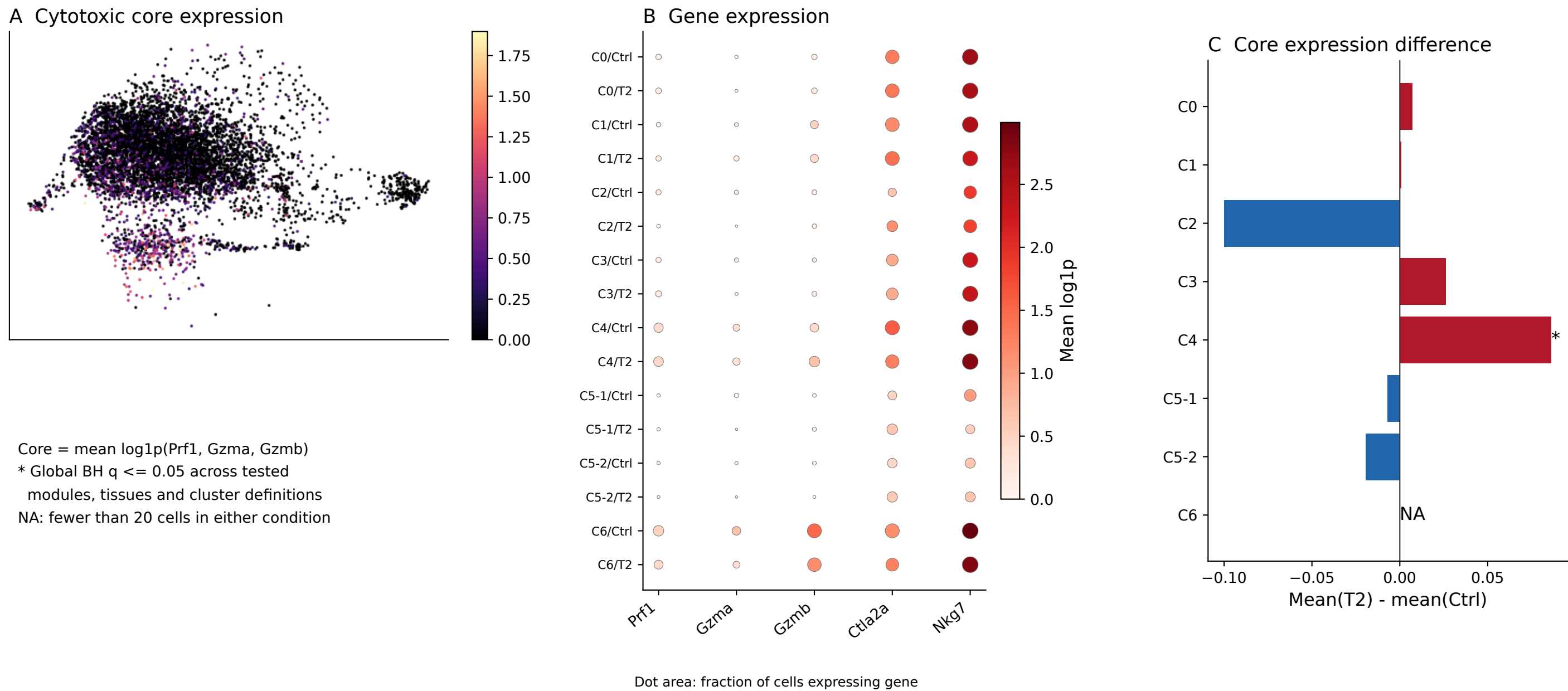
Primary endpoint: mean log-normalized expression of Prf1 / Gzma / Gzmb.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: cytotoxic expression and condition effects

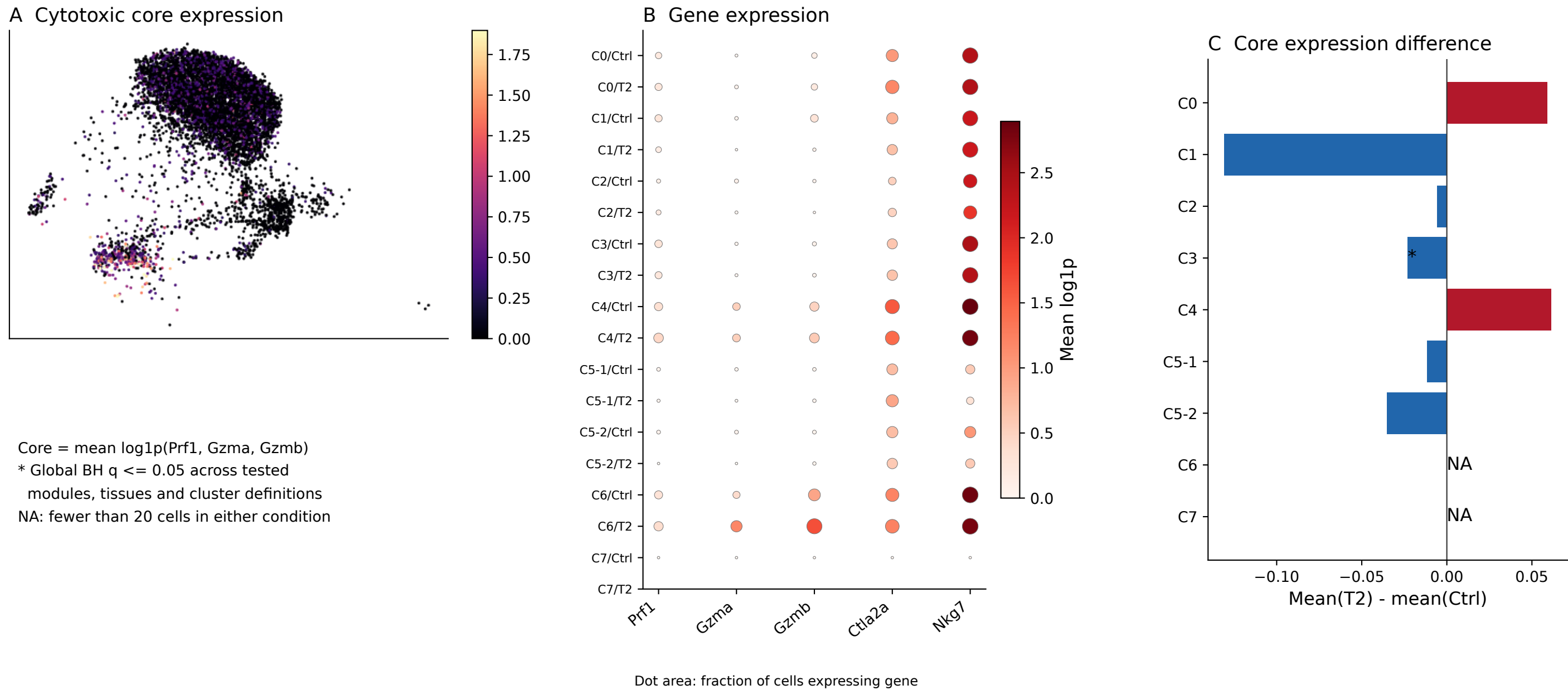
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Spleen: cytotoxic expression and condition effects

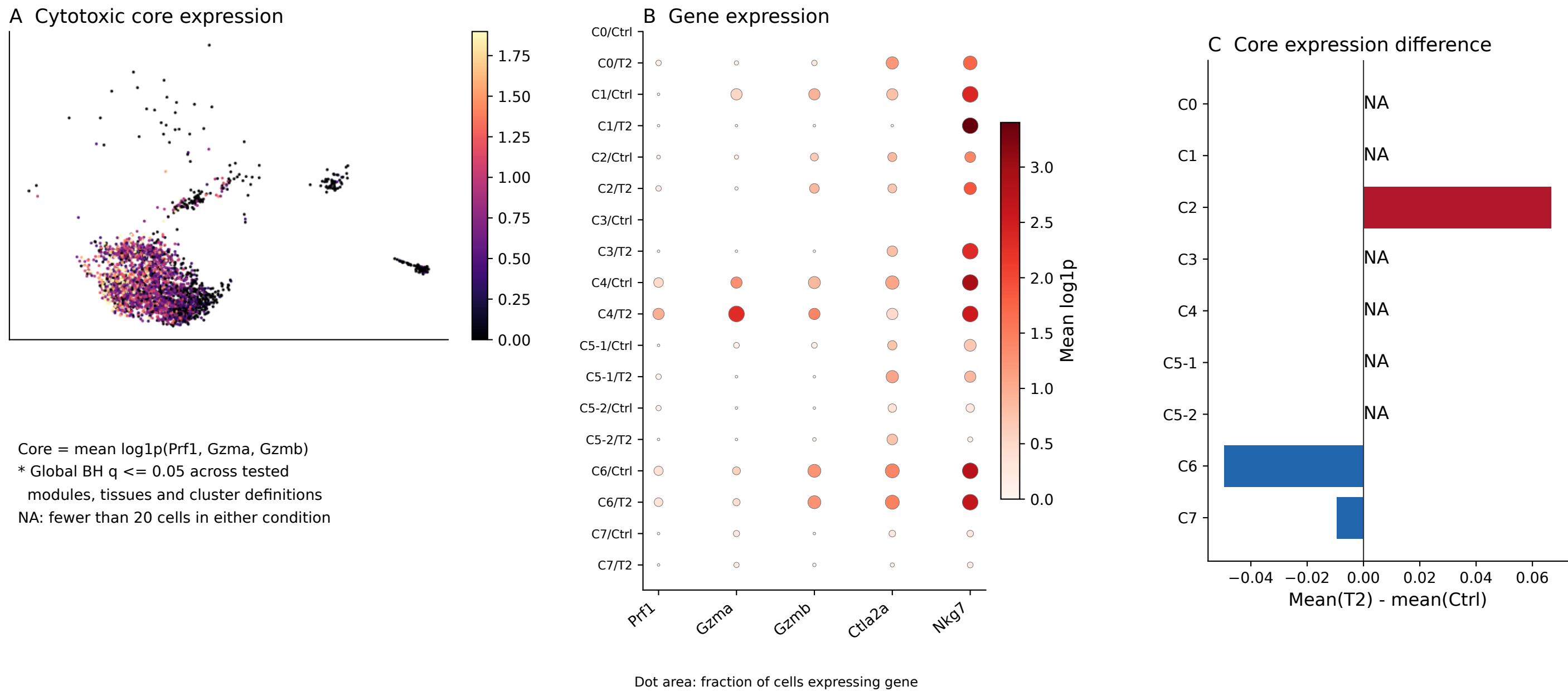
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Thymus: cytotoxic expression and condition effects

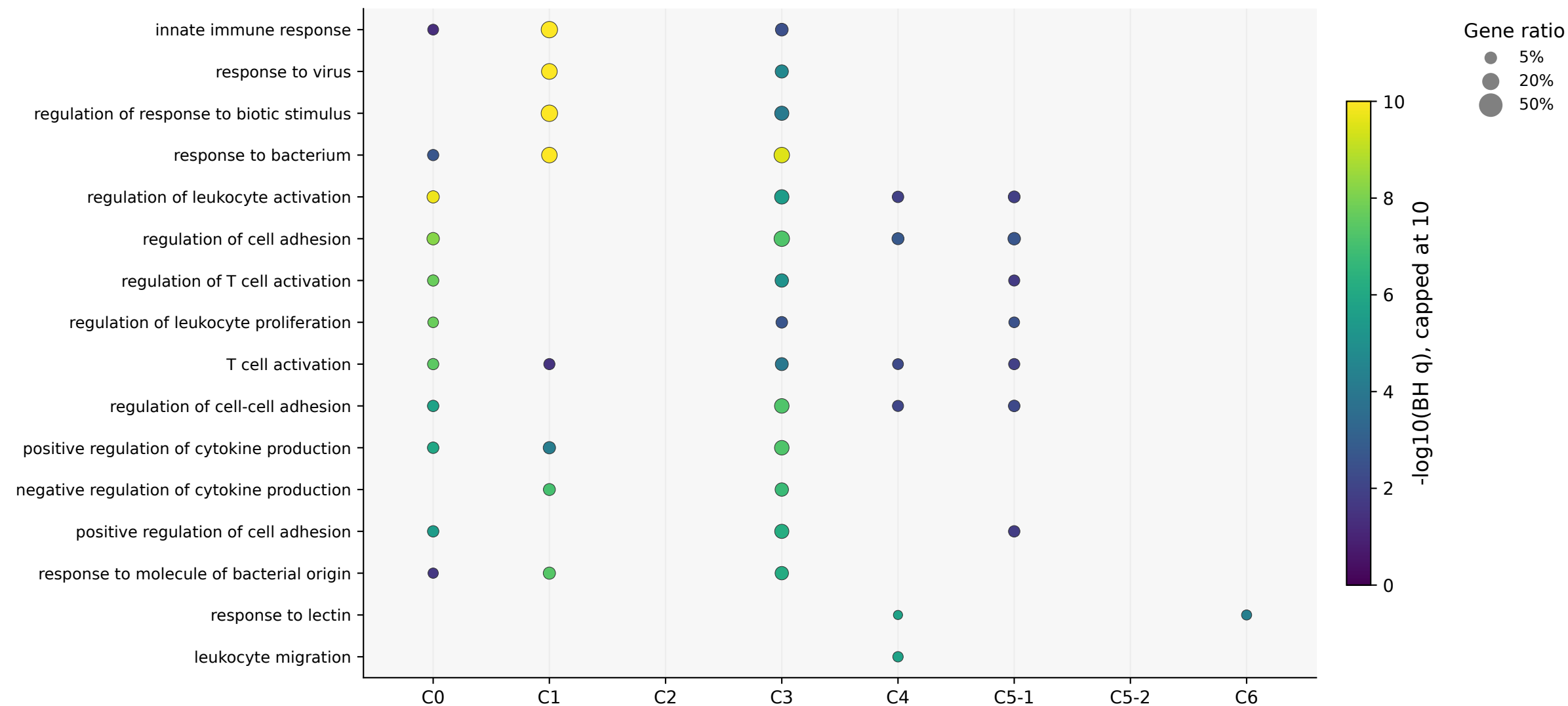
Original refined clusters; T2 - Ctrl within the same tissue and cluster.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

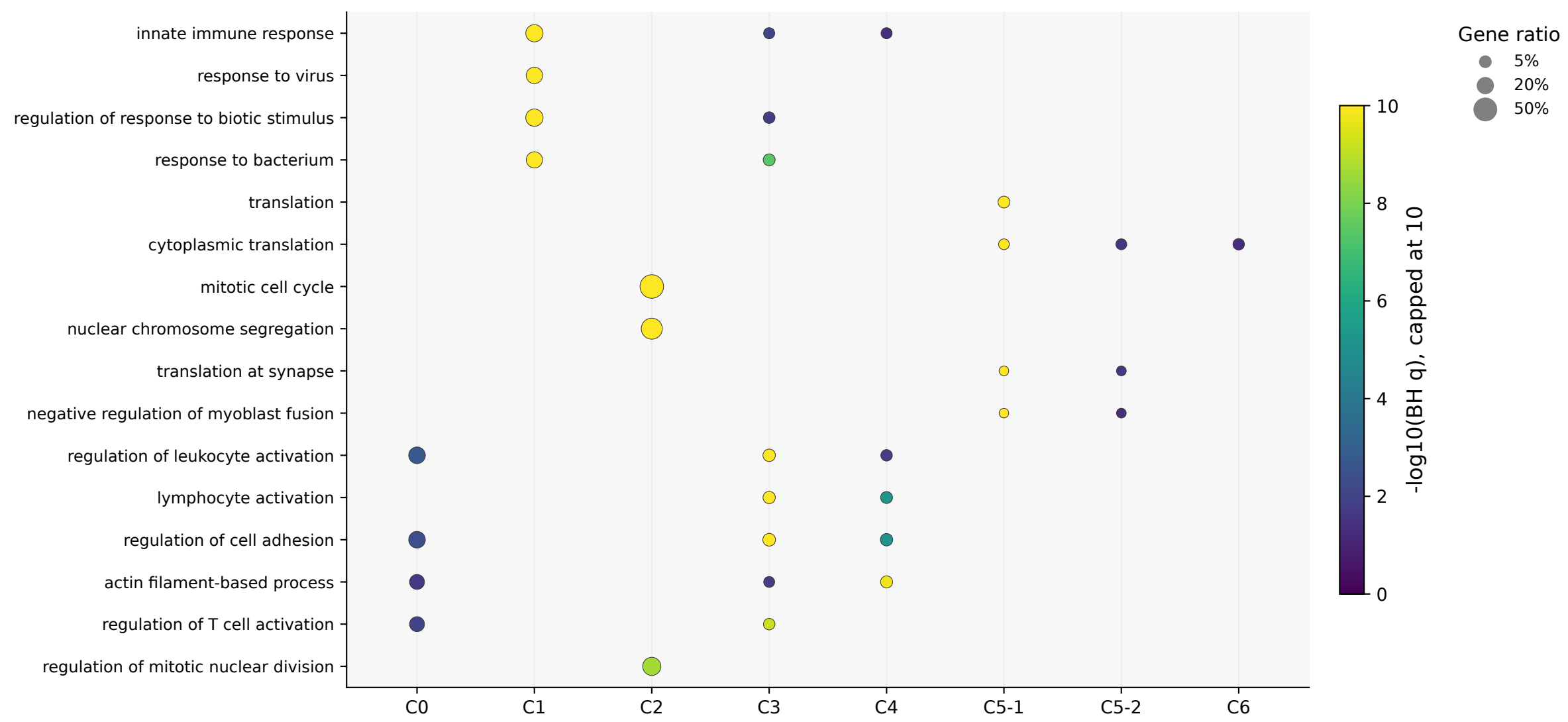


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

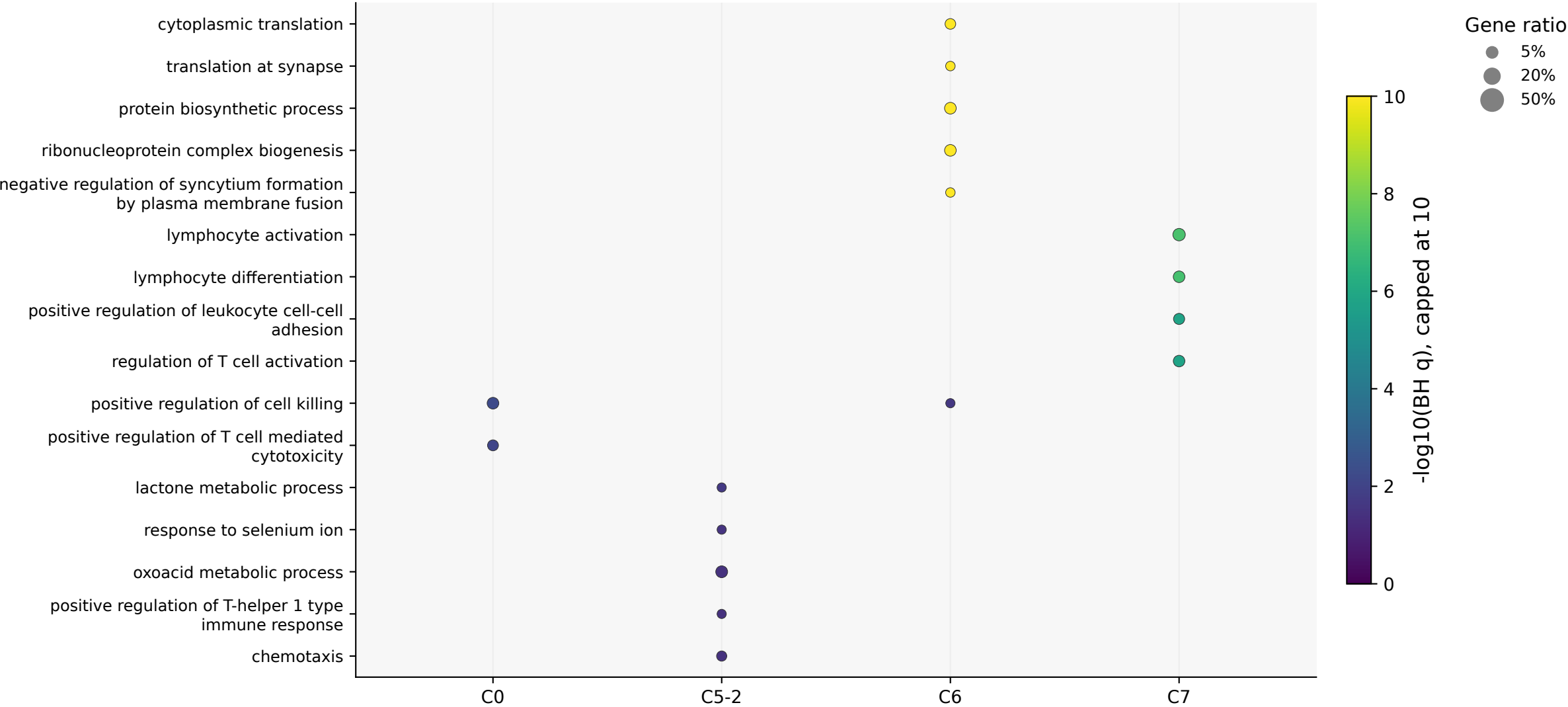


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: cluster identity

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

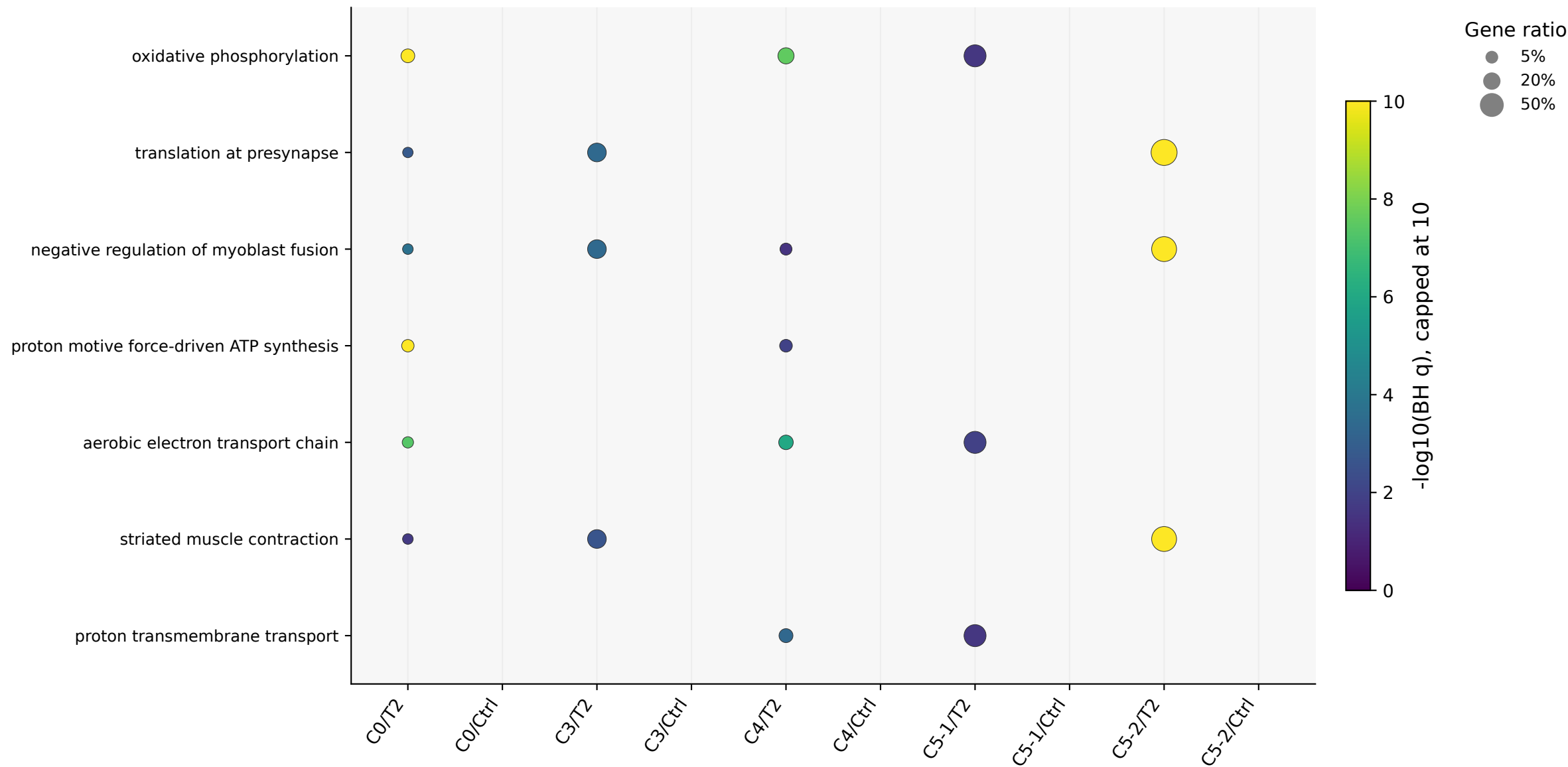


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Bone marrow: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

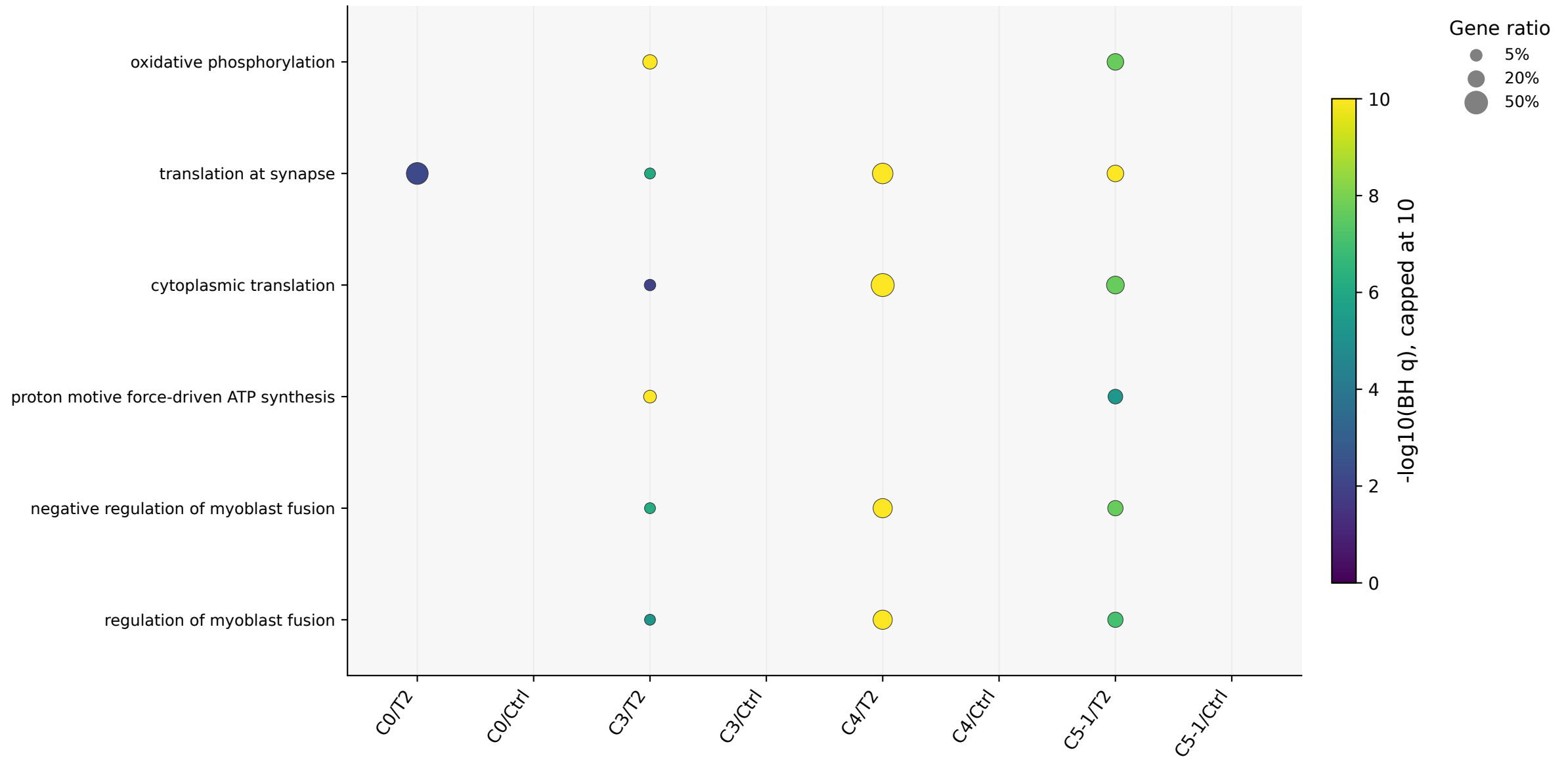


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; q <= .05; >=2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Spleen: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

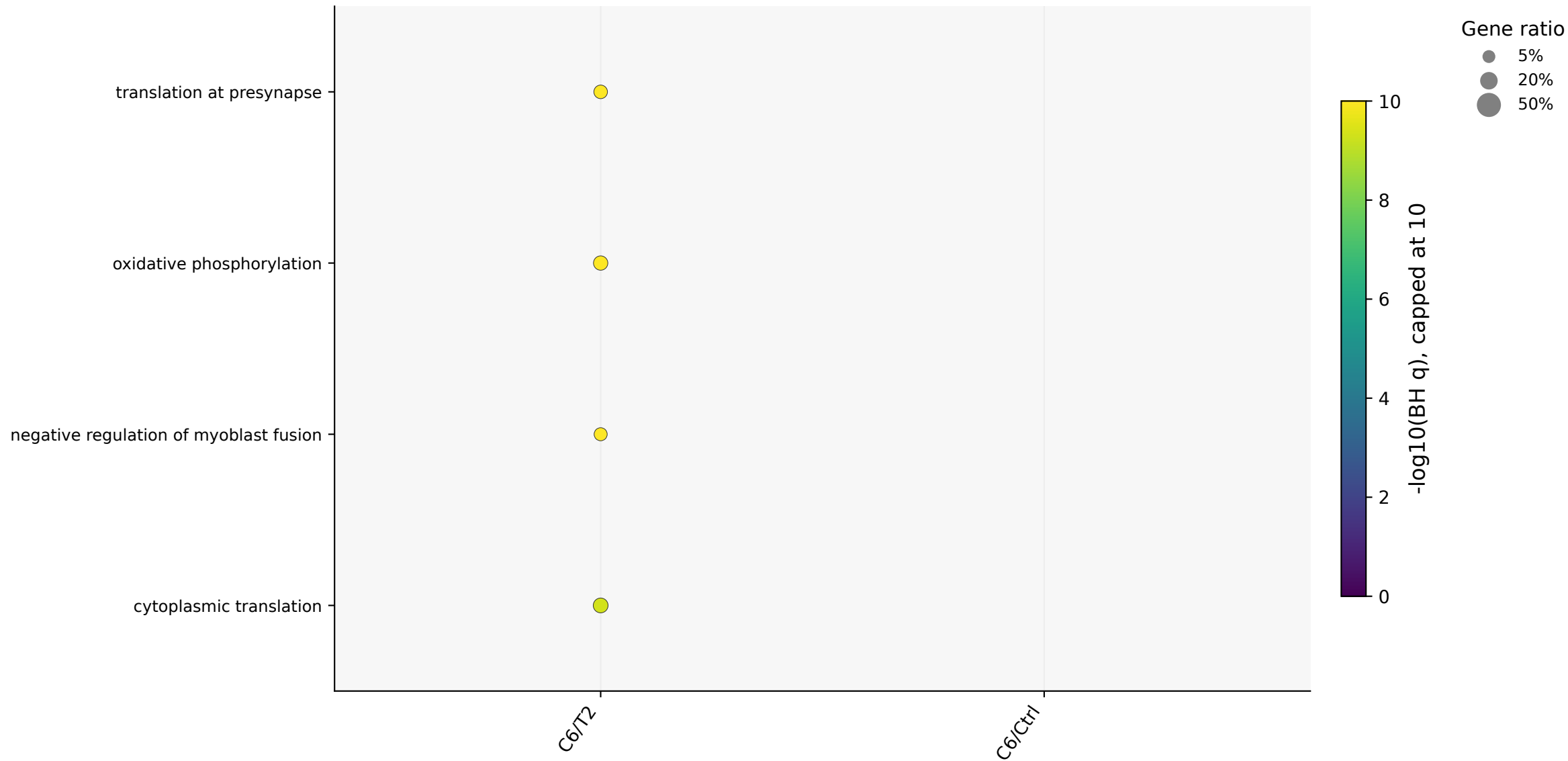


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; $q \leq .05$; ≥ 2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Thymus: GO biological processes: condition response

Within-tissue comparisons | dot area: query gene ratio | color: enrichment evidence

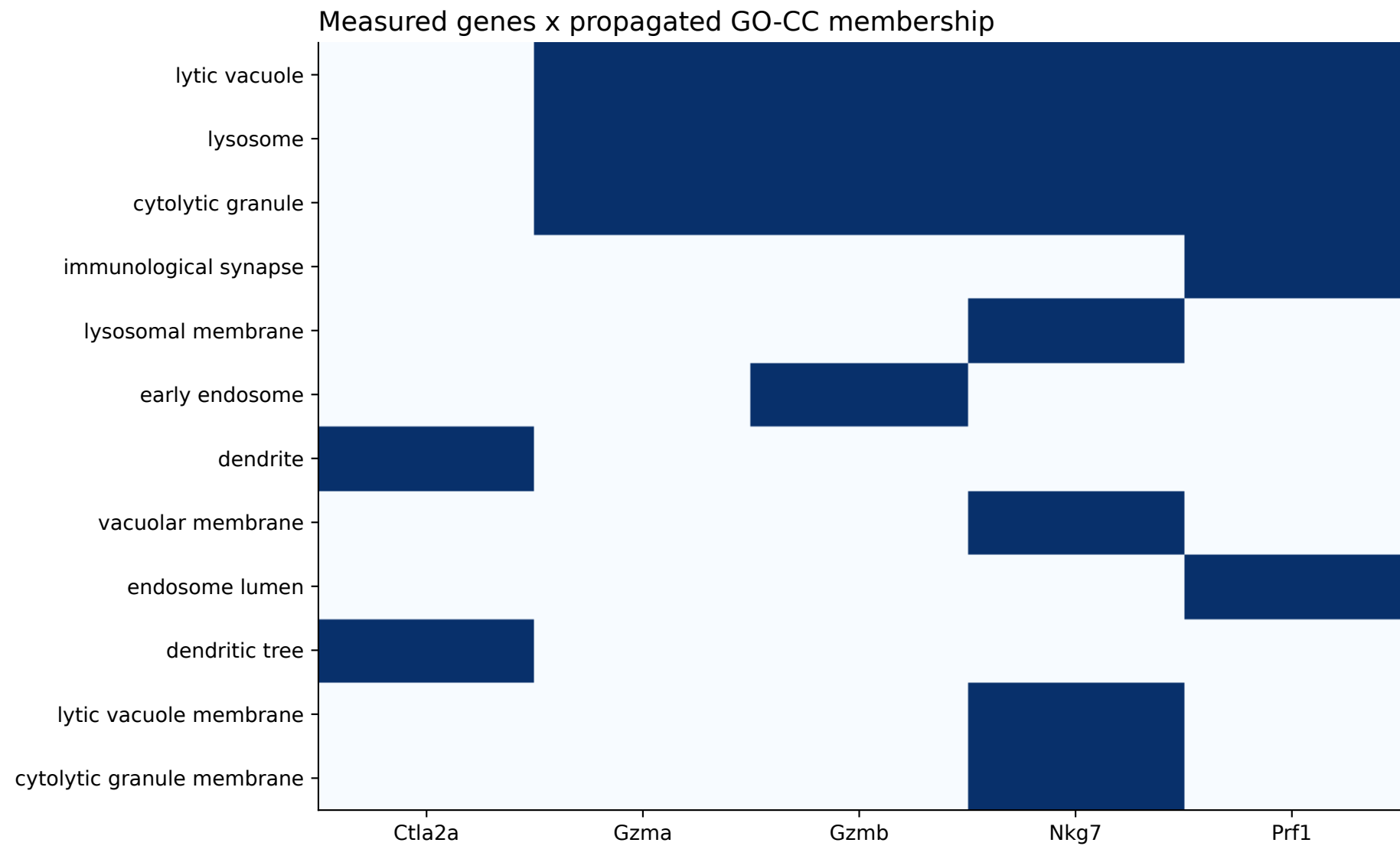


All BP/MF/CC tests retained. Only clusters with significant BP terms shown; q <= .05; >=2 hits; overlap redundancy reduced.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Blank = no displayed significant term, not proof of absence.

Cellular components of cytotoxic module genes

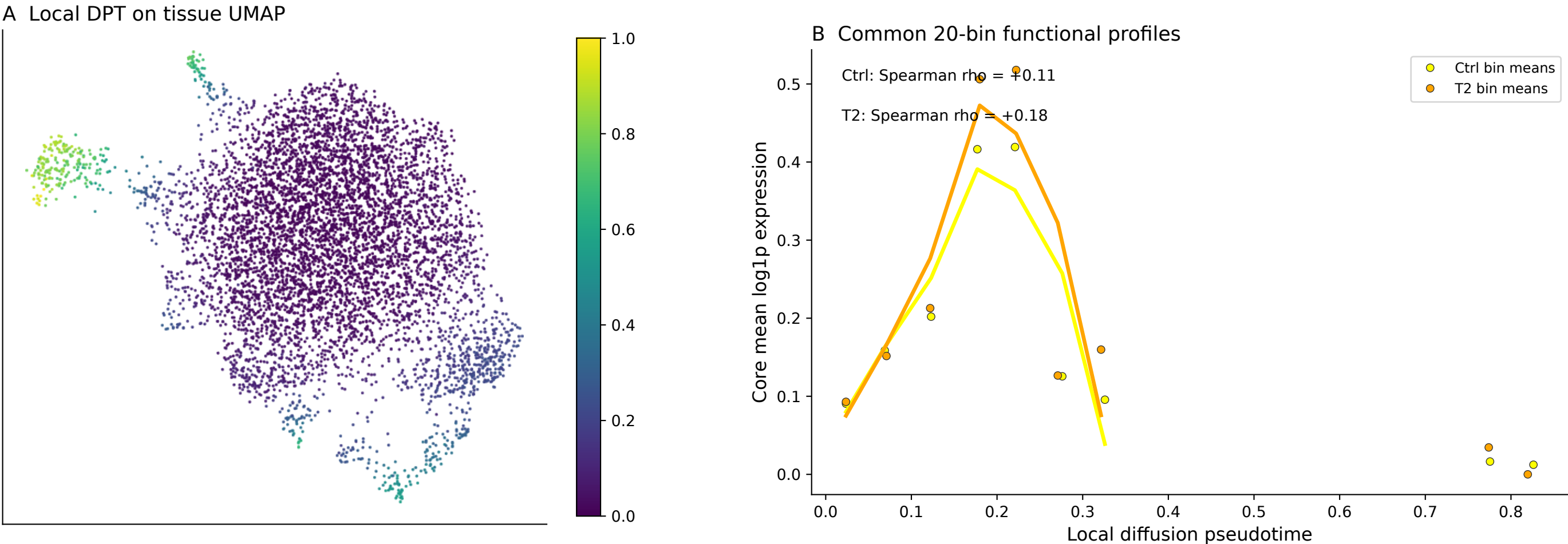
GO-CC membership supports localization; this is not a protein-interaction network.



Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Bone marrow: local cytotoxicity along pseudotime

PAGER-style functional curves on a newly computed within-tissue diffusion map.

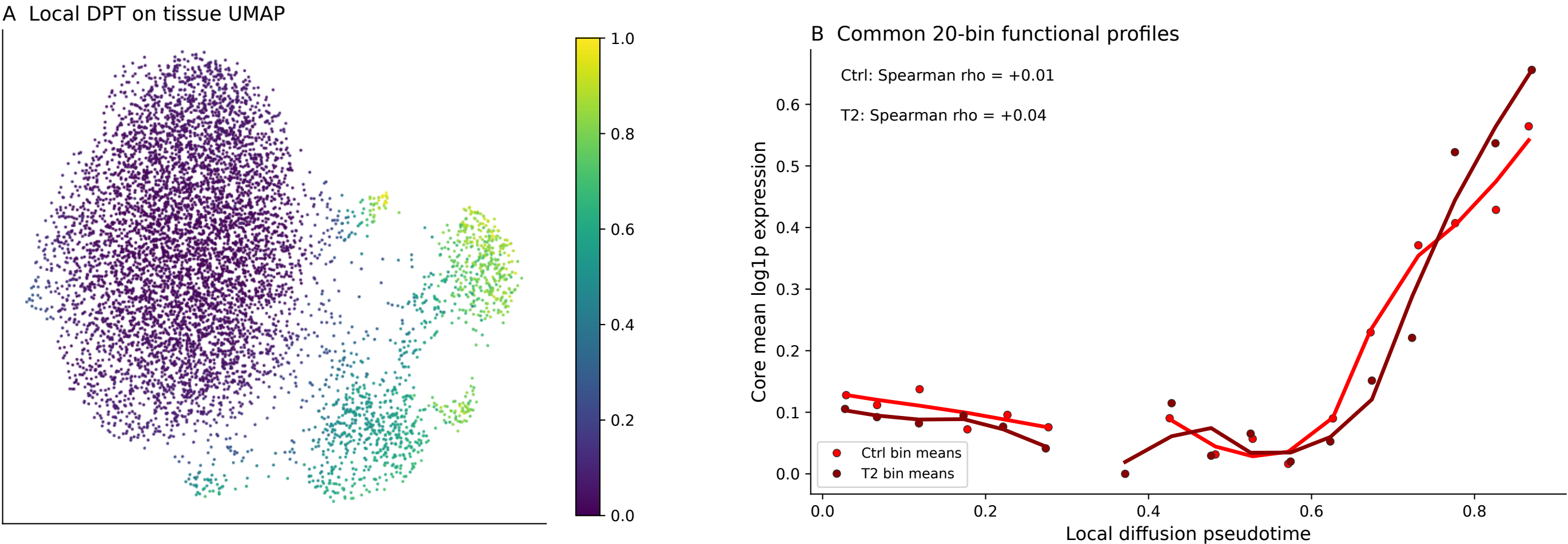


Root state: bone_marrow_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.89, 0.76.
Analyzed 6,110/6,110 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference. Pseudotime is an assumed ordering, not RNA velocity or lineage validation.

Spleen: local cytotoxicity along pseudotime

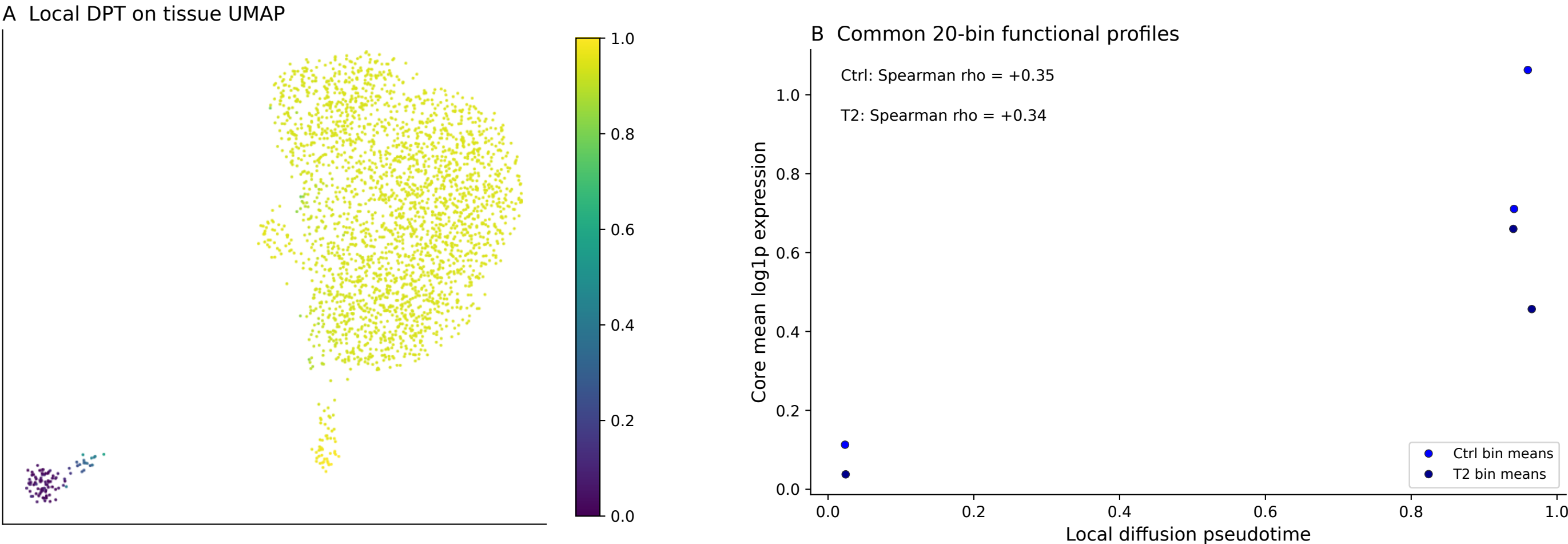
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: spleen_S0; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 0.97, 0.76.
Analyzed 6,793/6,793 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Thymus: local cytotoxicity along pseudotime

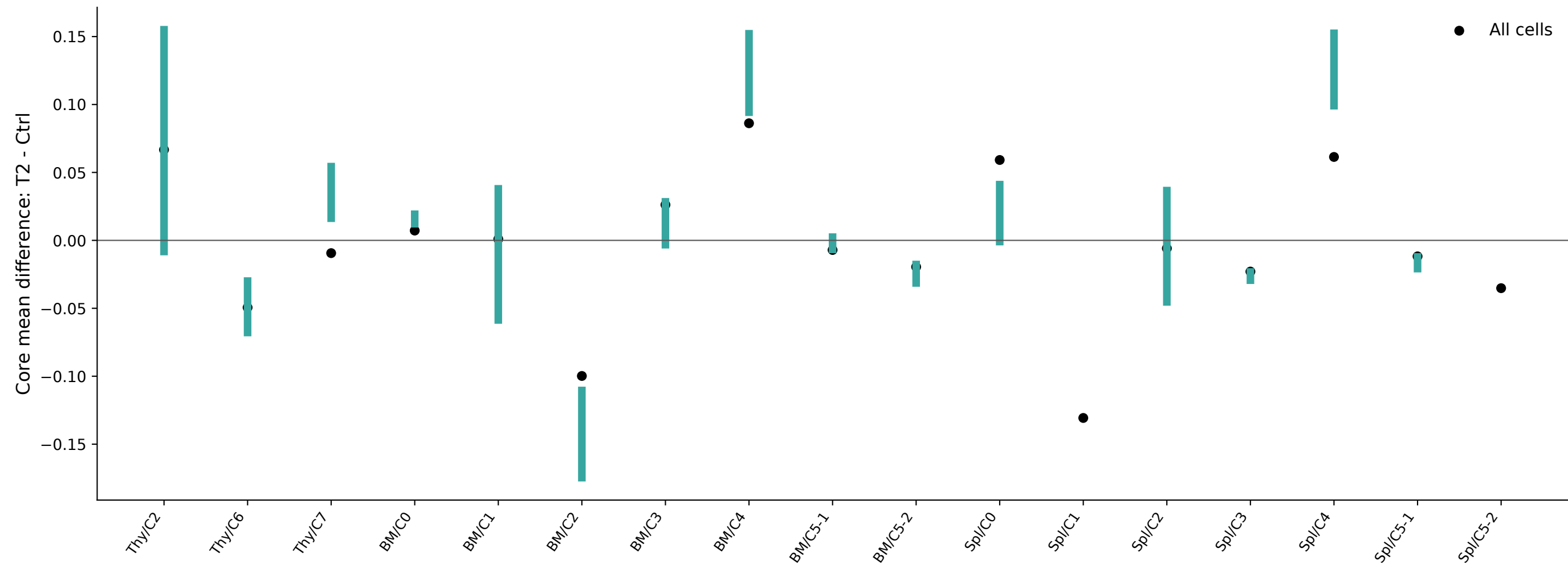
PAGER-style functional curves on a newly computed within-tissue diffusion map.



Root state: thymus_S5; measured root markers: Egr2 / Hivep3 (Cd24a unavailable). Root sensitivity rho: 1.00, 1.00.
Analyzed 2,629/2,629 cells; only bins with ≥ 20 cells shown. Smooth lines are descriptive.

Robustness: QC matching and alternative clustering

Five condition-balanced QC resamples within each tissue x cluster.

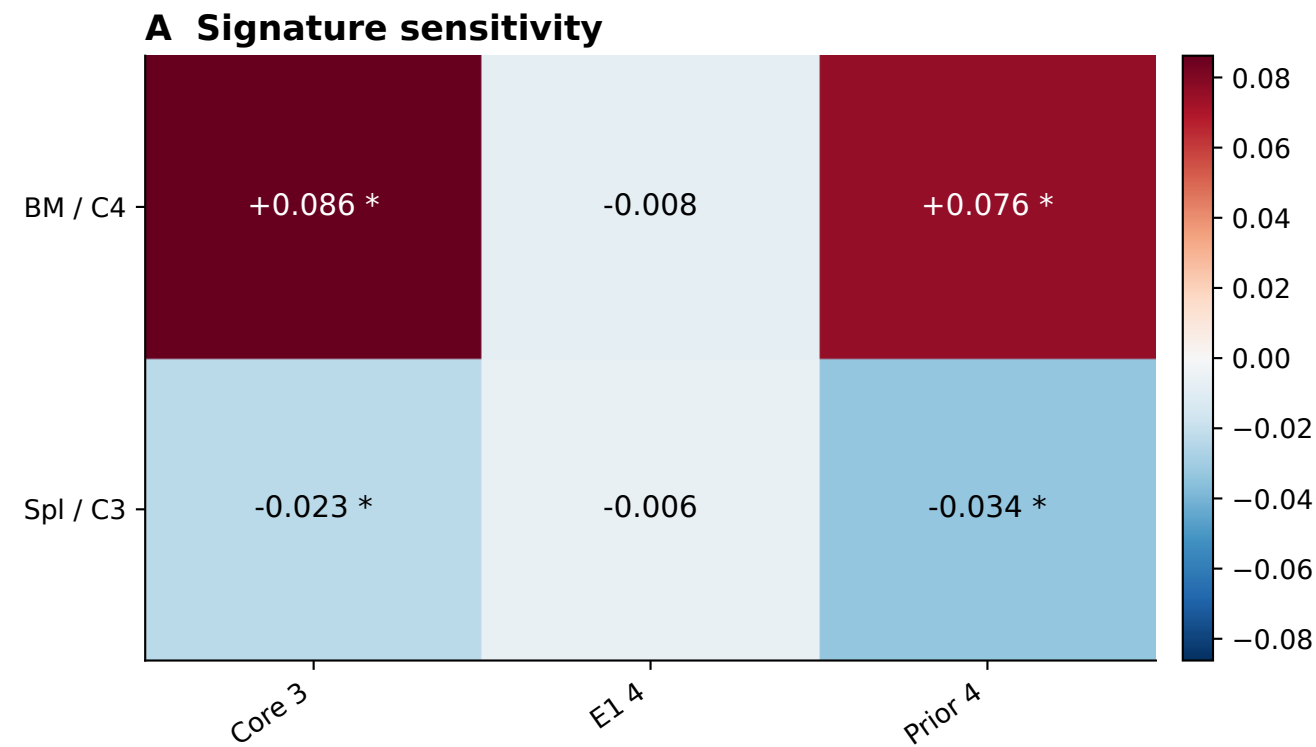


Teal range: five QC-matched effect estimates (not a confidence interval).
Stable tissue clusters are a separate sensitivity analysis; their labels are not interchangeable with original C labels.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Core signature changes do not imply a full killing pathway

Two reproducible local core signals; the broader E1 signature is not significant.



B Complementary evidence from the same dataset

Bone marrow C4 core: +0.086; q = 0.025
Spleen C3 core: -0.023; q = 0.0015
Both: same direction in all 5 QC resamples

Stable-cluster effects: +0.111 and -0.022
Original-cell overlap: 98.7% and 99.5%

E1 (+Ctla2a): q = 0.93 and 0.53
Killing-related GO terms: no q ≤ 0.05

The evidence supports a localized core-effector transcriptional shift.
It does not establish increased/decreased killing activity or coordinated activation of a full cytotoxic pathway.

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Interpretation: localized effector changes with shared metabolic signals

Separate direct observations, working hypotheses and unresolved evidence

Population / scope	Observation	Interpretation
Bone marrow C4	Core score +0.086; q=0.025. Gzmb detection 29% to 44%.	Selective Gzmb-associated effector transcriptional shift.
Spleen C3	Core score -0.023; q=0.0015. Prf1 and Nkg7 decrease.	Small reduction in effector-associated transcripts.
Both populations	T2-up genes enriched for oxidative phosphorylation.	Respiration-related gene expression shifts; no metabolic flux measured.
Limits	Broader E1 signature and killing-related GO are not significant.	No demonstrated change in killing capacity or animal-level treatment effect.

Both core effects persist under QC matching and overlapping stable clusters.
These same-data checks support candidates; independent animals and functional assays remain necessary.

iNKT: results in the reference presentation formats

Existing analyses from July–September 2026 • presentation revision: 15 September 2026

Analysis	Reference	Presentation used here
Sample/QC, markers, DEG and pathway pages	Original iNKT deck	Counts + plots; gene dotplots + numerical tables
Cluster composition	Blood Fig. 3C	Two 100% stacked-bar plots, each with its own denominator
Differential expression	Blood Fig. 3D	Volcano: log2FC vs –log10 nominal P; blue/red direction
Up-DEG pathway enrichment	Blood Fig. 3E	Single score column, –ln(P), with driver-gene callouts
Shared upregulated genes	Blood Fig. 3F	Four-set Venn; identical saved membership and exact counts
Signed GSEA; batch/state sensitivity	Additional analyses	Separate figures; NES is not the paper enrichment score

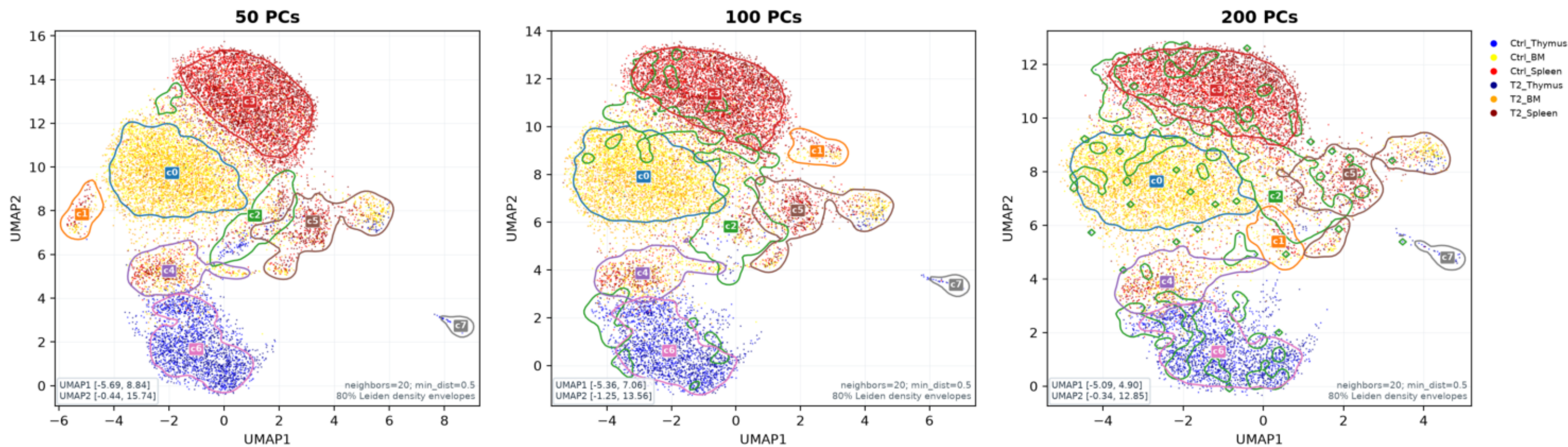
Our cells are iNKT; the Blood paper studies NK cells. Current C# labels are not paper K# or legacy c# identities.

Same analytical question → same figure grammar. Different algorithms, populations and thresholds are stated explicitly.

UMAP: 50, 100 and 200 principal components

Controlled sensitivity comparison; selected 50 PCs

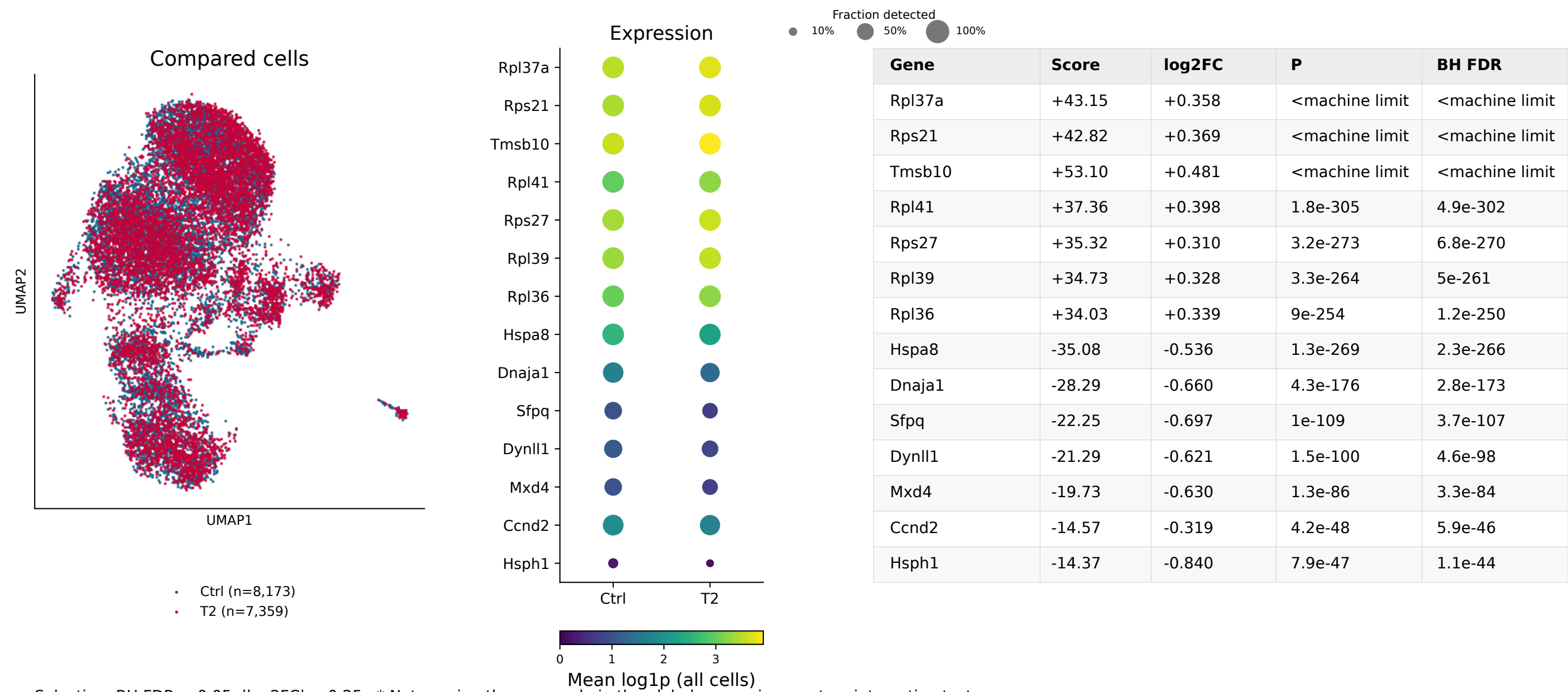
UMAP PC sensitivity — only n_pcs changes; fixed current Leiden labels



PC number varies; all other settings are fixed. Labels are fixed reference clusters, not separate reclustering outcomes.

T2 vs. Control DEGs: global

Expression/detection dotplot alongside the exact same genes in a signed DEG table

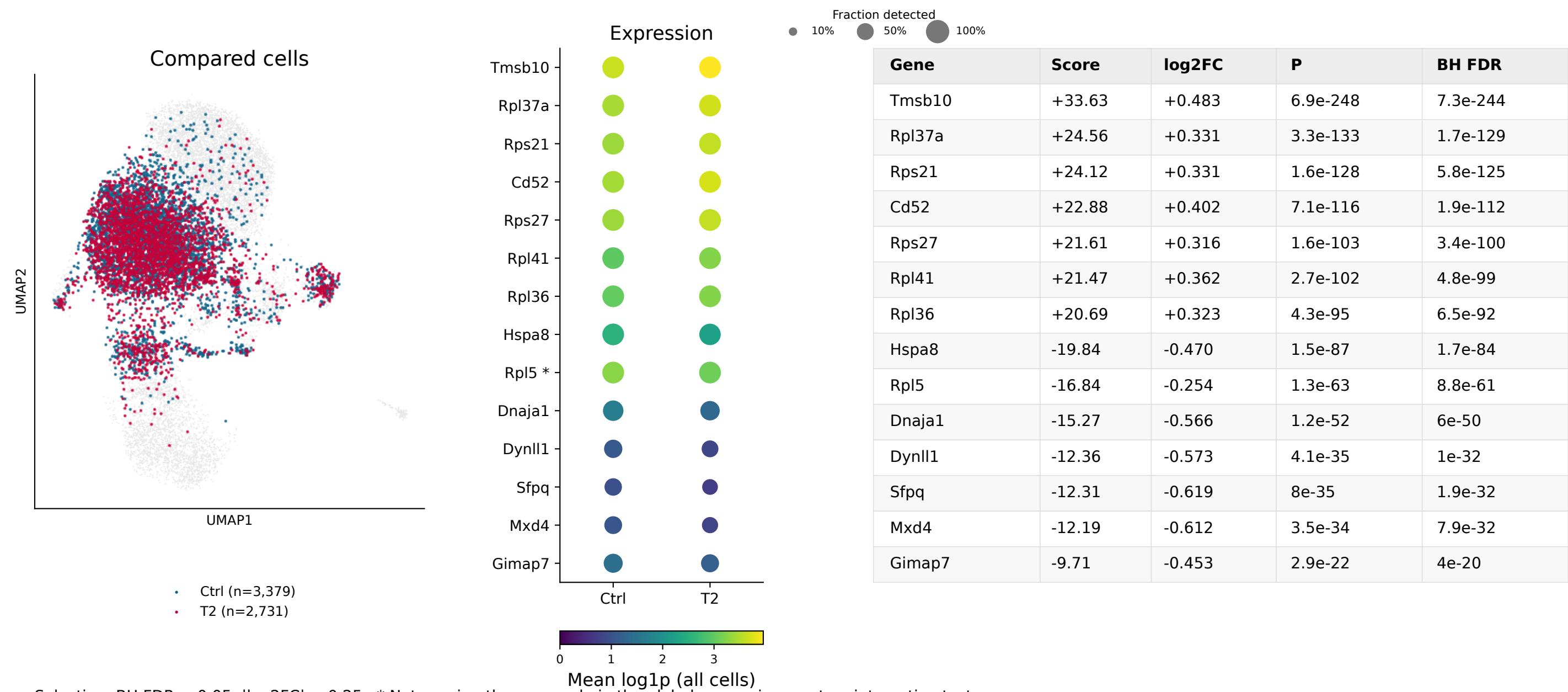


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: BM

Expression/detection dotplot alongside the exact same genes in a signed DEG table

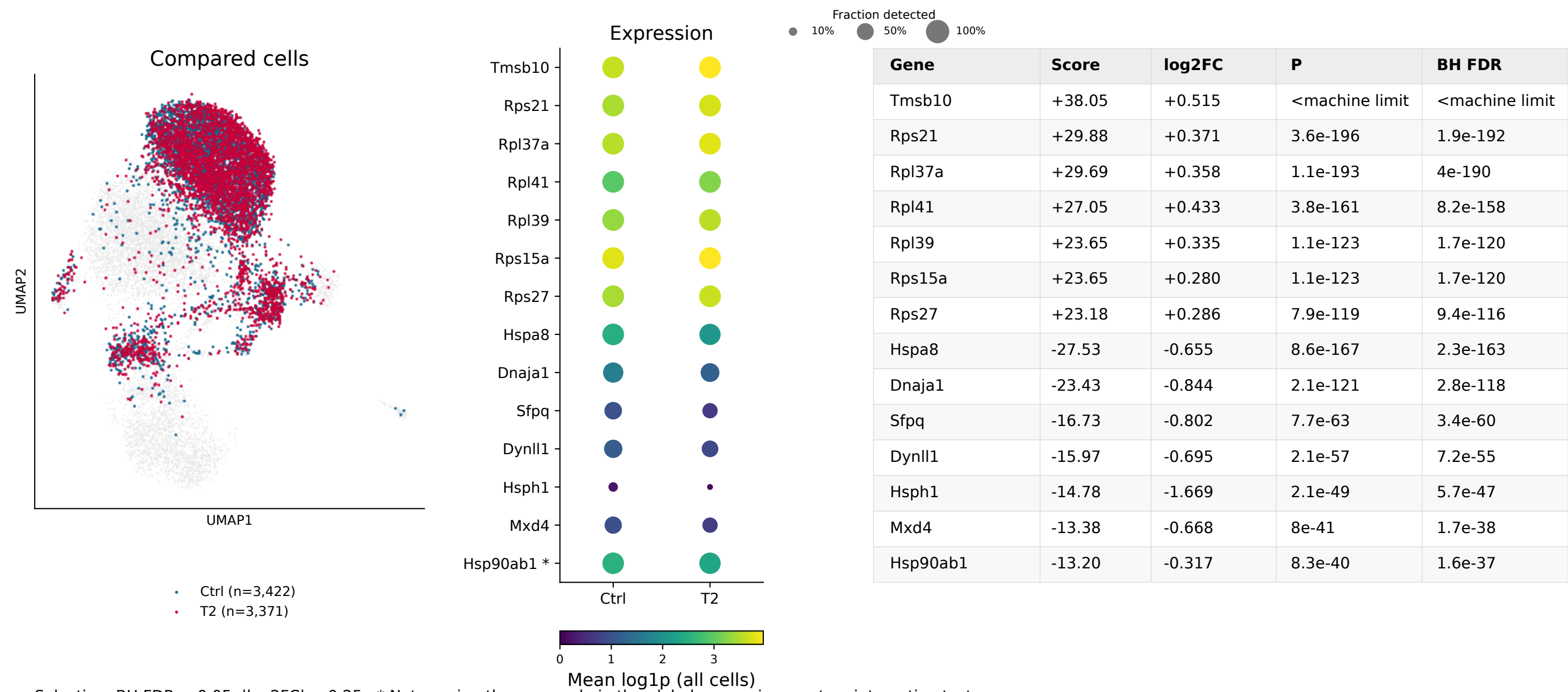


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: Spl

Expression/detection dotplot alongside the exact same genes in a signed DEG table

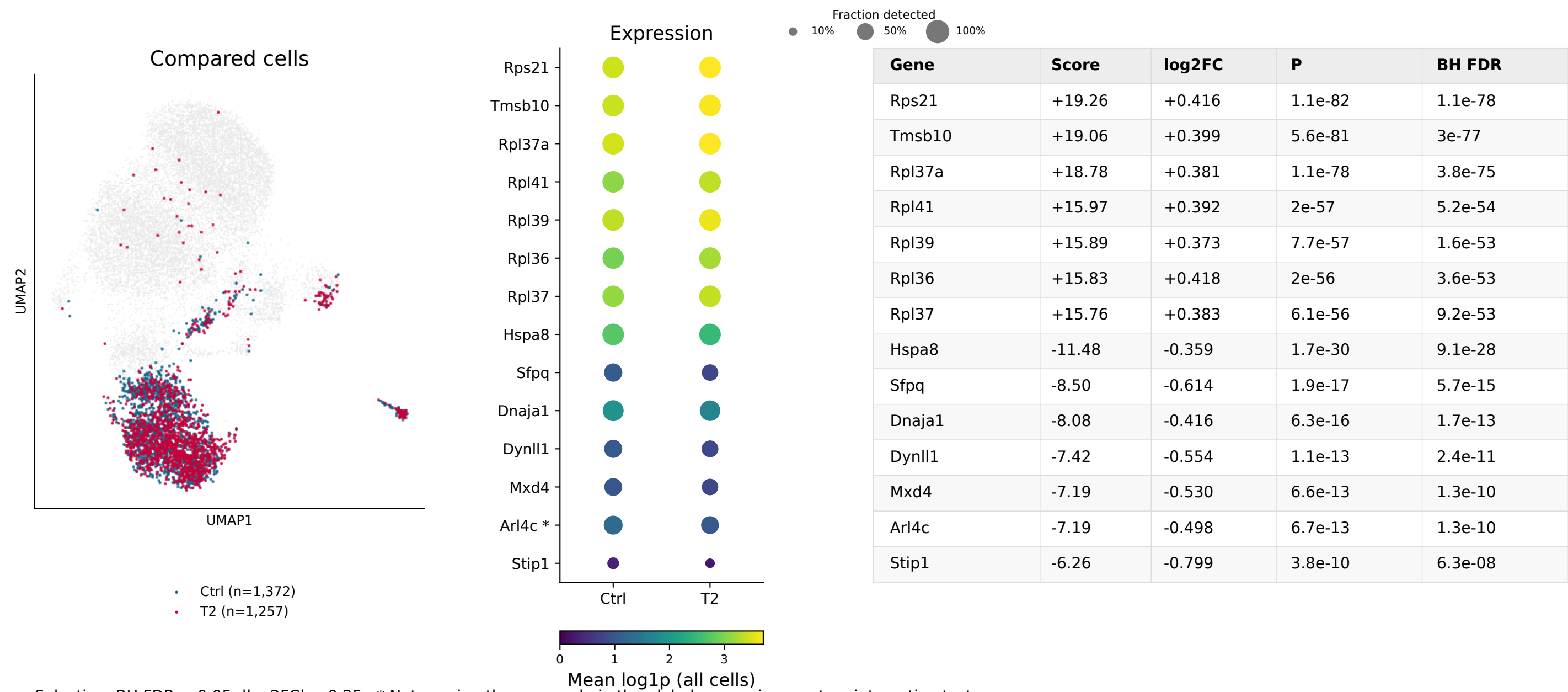


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: Thy

Expression/detection dotplot alongside the exact same genes in a signed DEG table

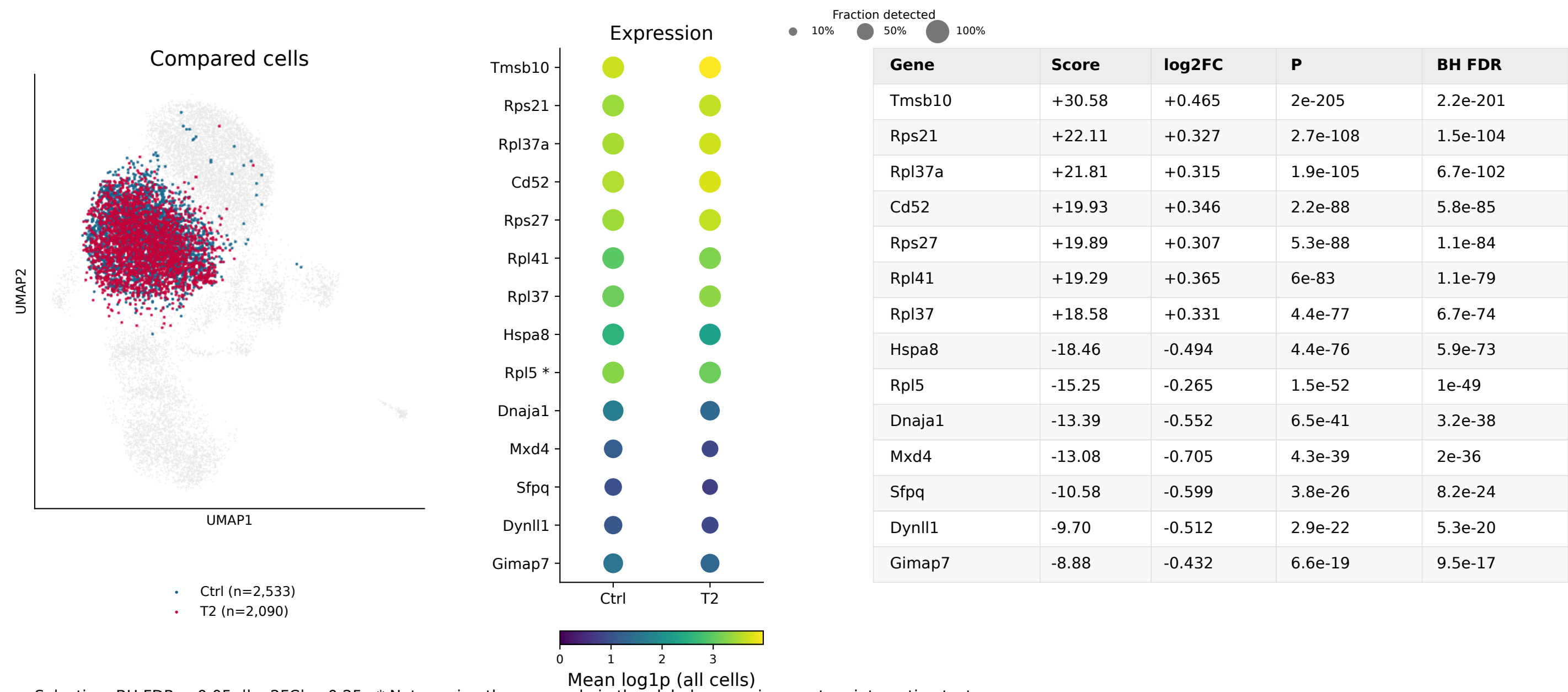


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: C0 / BM

Expression/detection dotplot alongside the exact same genes in a signed DEG table

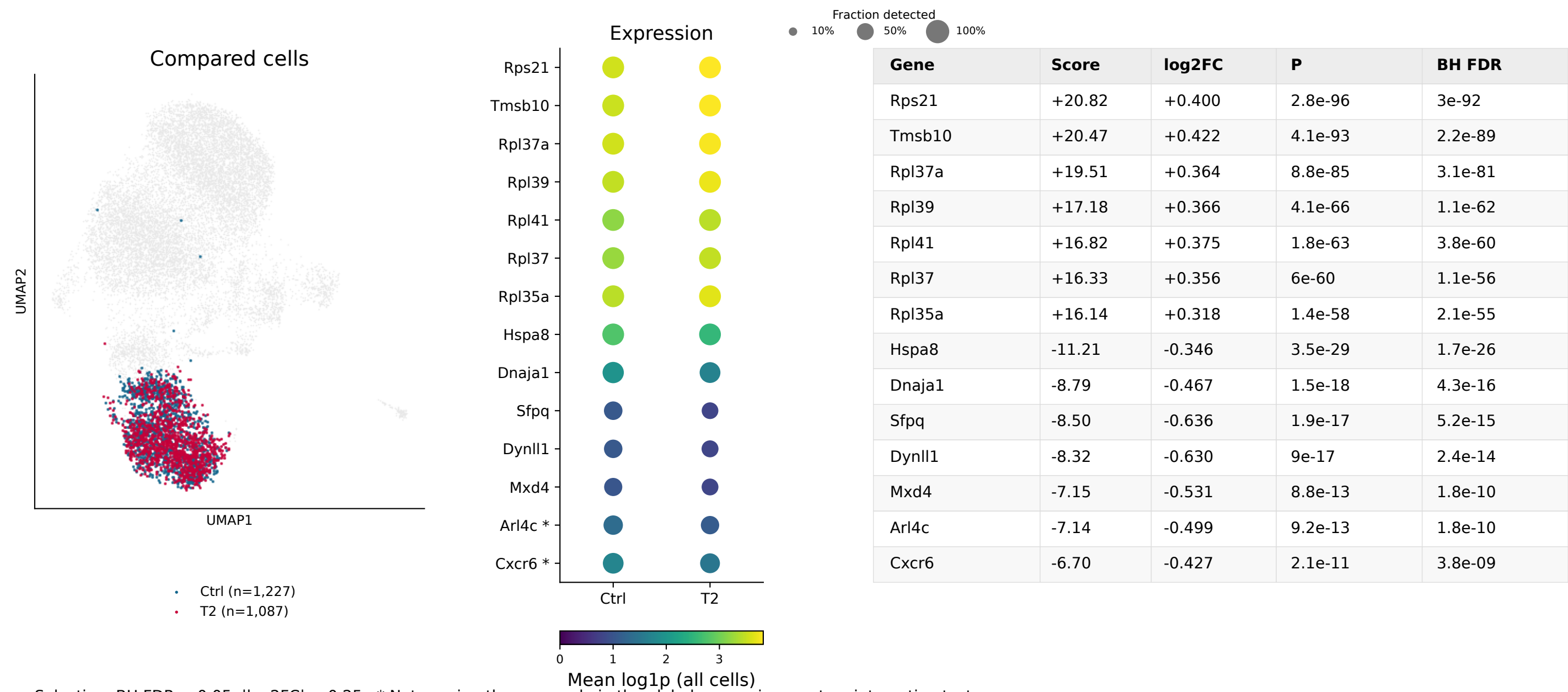


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: C6 / Thy

Expression/detection dotplot alongside the exact same genes in a signed DEG table

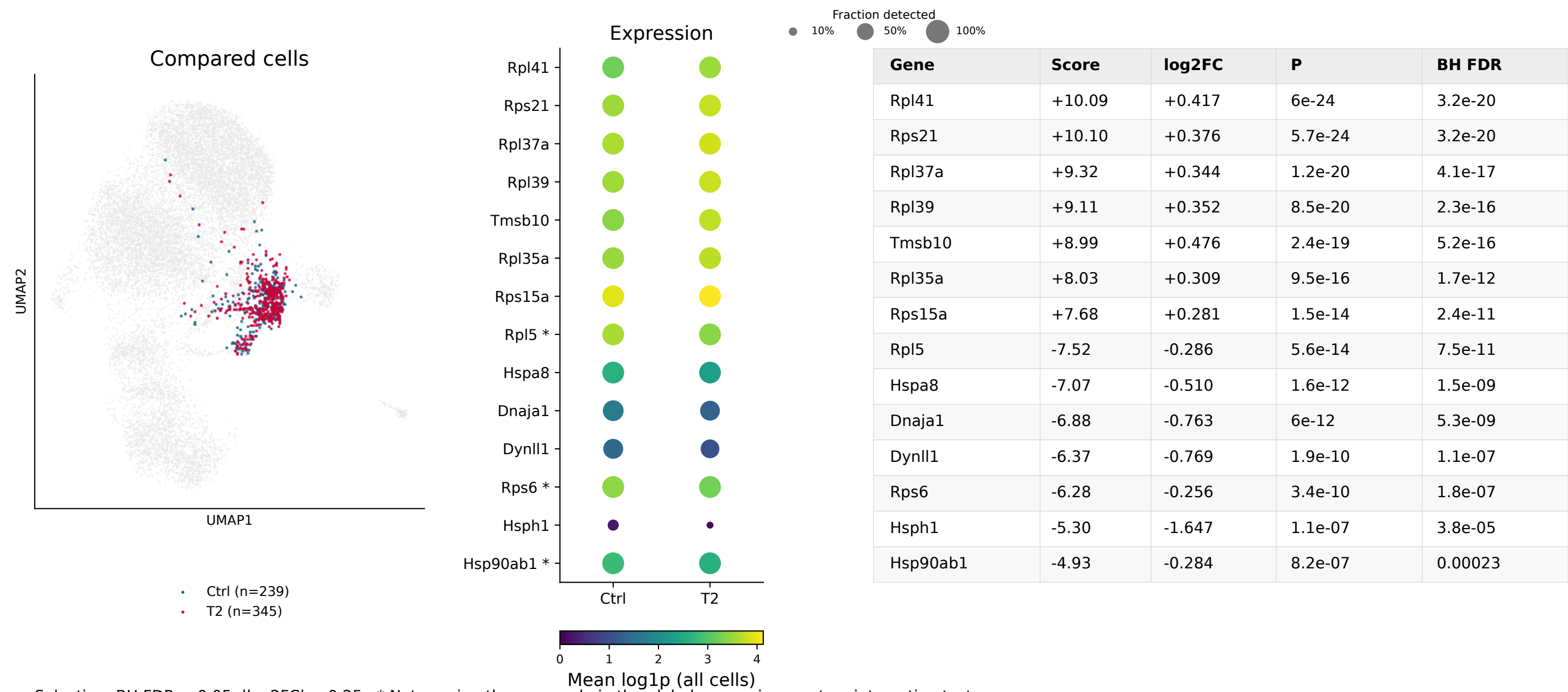


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: C5-1 / Spl

Expression/detection dotplot alongside the exact same genes in a signed DEG table

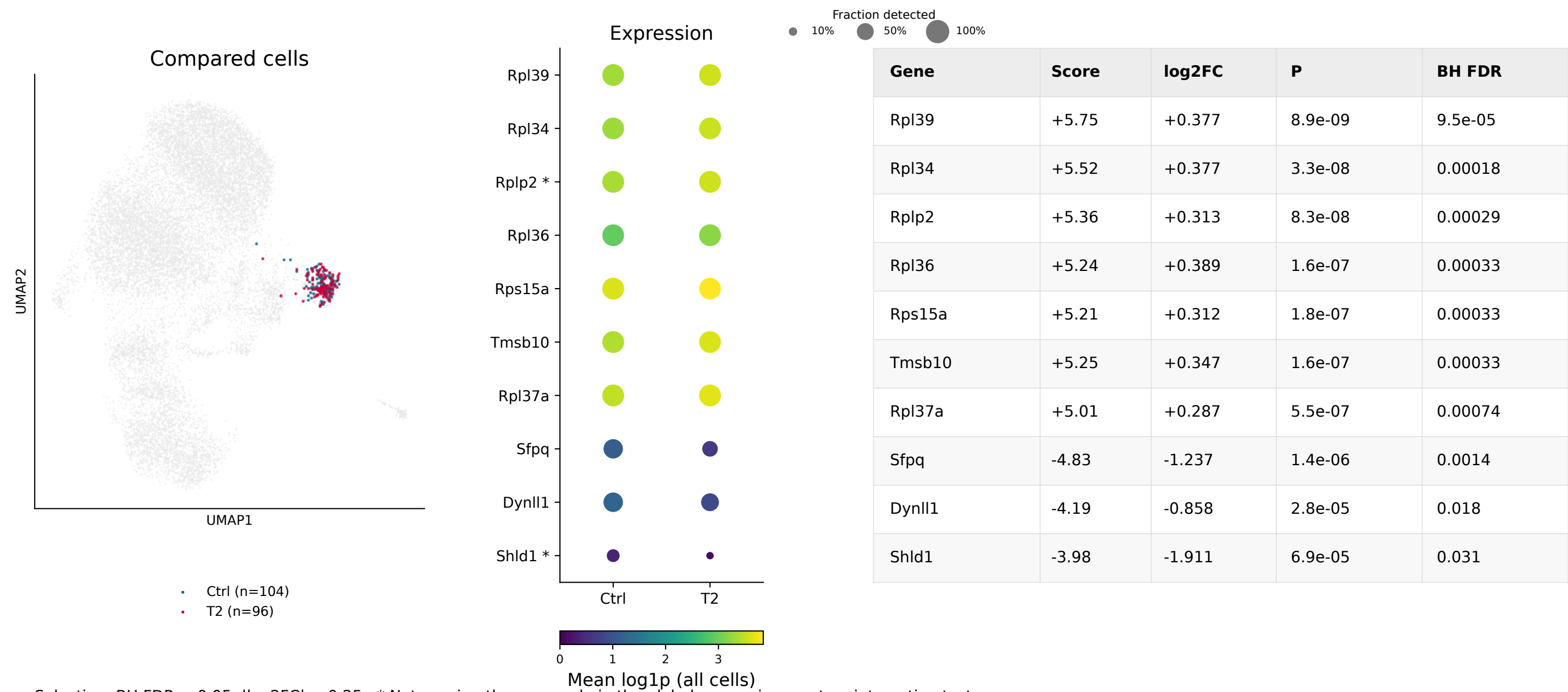


Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

T2 vs. Control DEGs: C5-2 / BM

Expression/detection dotplot alongside the exact same genes in a signed DEG table



Selection: BH FDR ≤ 0.05, |log2FC| ≥ 0.25. * Not passing the same rule in the global comparison; not an interaction test.

Displayed: 7 strongest FDR-ranked genes per direction, when available; full gene table linked in the source index.

Pathways: C6 / Thy

Legacy c1_thymus pathway vocabulary; current full-member ORA with explicit up/down direction

Pathway / reaction	Legacy source	Old FDR	T2-up FDR	Ctrl-up FDR	Current overlapping genes
hsa05020 Prion disease - Homo sapiens (human)	KEGG 2021 HUMAN	5.9e-05	2.6e-10	0.44	T2: Cox5b, Cox6c, Cox8a +16 more Ctrl: Bax, Ccl5, Csnk2a2 +6 more
p23 (PTGES3) binds HSP90:ATP:FKBP5:nascent protein	Reactome 2021	4.2e-06	0.0047	0.00066	T2: Hspa1a, Hspa1b Ctrl: Hspa8, Ptges3, Stip1
p23 (PTGES3) binds HSP90:ATP:FKBP4:nascent protein	Reactome 2021	4.2e-06	0.0047	0.00066	T2: Hspa1a, Hspa1b Ctrl: Hspa8, Ptges3, Stip1
HSP40s activate intrinsic ATPase activity of HSP70s in the cytosol	Reactome 2021	2.8e-05	0.0018	0.00066	T2: Dnajb1, Hspa1a, Hspa1b Ctrl: Dnajc2, Hspa14, Hspa5 +1 more
HSP40s activate intrinsic ATPase activity of HSP70s in the nucleoplasm	Reactome 2021	3.3e-05	0.0018	0.00066	T2: Dnajb1, Hspa1a, Hspa1b Ctrl: Dnajc2, Hspa14, Hspa5 +1 more
STIP1(HOP) binds HSP90 and HSP70:HSP40:nascent protein	Reactome 2021	7.2e-09	0.0013	0.0012	T2: Dnajb1, Hspa1a, Hspa1b Ctrl: Dnaja1, Hspa8, Stip1
ATP hydrolysis by HSP70	Reactome 2021	1.9e-09	0.0013	0.011	T2: Dnajb1, Hspa1a, Hspa1b Ctrl: Dnaja1, Hspa8
HSP70 binds to HSP40:nascent protein	Reactome 2021	1.8e-09	0.0013	0.011	T2: Dnajb1, Hspa1a, Hspa1b Ctrl: Dnaja1, Hspa8

Shown: 8 strongest current min(up-FDR, down-FDR), retaining original names; first 3 drivers per direction shown; complete records in exported table.

Legacy/current cluster correspondence is a response comparison, not an identity claim; pathway versions and thresholds differ.

Pathways: C0 / BM

Legacy c5_bone_marrow pathway vocabulary; current full-member ORA with explicit up/down direction

Pathway / reaction	Legacy source	Old FDR	T2-up FDR	Ctrl-up FDR	Current overlapping genes
hsa05020 Prion disease - Homo sapiens (human)	KEGG 2021 HUMAN	3.9e-11	5.3e-13	1	T2: Atp5j, Cox5b, Cox6b1 +26 more Ctrl: Hspa8, Psmd2, Psmd6 +3 more
FKBP4 binds HSP90:ATP:STIP1:HSP70:nascent protein	Reactome 2021	3e-05	0.046	0.0062	T2: Hspa1a, Hspa1b Ctrl: Fkbp4, Hspa8, Stip1
HSF1 acetylation at Lys80	Reactome 2021	3e-05	0.067	0.0062	T2: Hspa1a, Hspa1b Ctrl: Ep300, Fkbp4, Hspa8 +1 more
p23 (PTGES3) binds HSP90:ATP:FKBP4:nascent protein	Reactome 2021	4.3e-05	0.046	0.0062	T2: Hspa1a, Hspa1b Ctrl: Fkbp4, Hspa8, Stip1
STIP1(HOP) binds HSP90 and HSP70:HSP40:nascent protein	Reactome 2021	1.6e-07	0.051	0.011	T2: Hspa1a, Hspa1b Ctrl: Dnaja1, Hspa8, Stip1
Acetylated HSF1 dissociates from DNA	Reactome 2021	1.3e-05	0.062	0.015	T2: Hspa1a, Hspa1b Ctrl: Fkbp4, Hspa8, Hsph1
HSP70:DNAJB1 binds HSF1	Reactome 2021	1.3e-05	0.062	0.015	T2: Hspa1a, Hspa1b Ctrl: Fkbp4, Hspa8, Hsph1
HSF1-mediated gene expression	Reactome 2021	5.9e-05	0.062	0.015	T2: Hspa1a, Hspa1b Ctrl: Fkbp4, Hspa8, Hsph1

Shown: 8 strongest current min(up-FDR, down-FDR), retaining original names; first 3 drivers per direction shown; complete records in exported table.

Legacy/current cluster correspondence is a response comparison, not an identity claim; pathway versions and thresholds differ.

Pathways: C5-1 / Spl

Legacy c6_spleen pathway vocabulary; current full-member ORA with explicit up/down direction

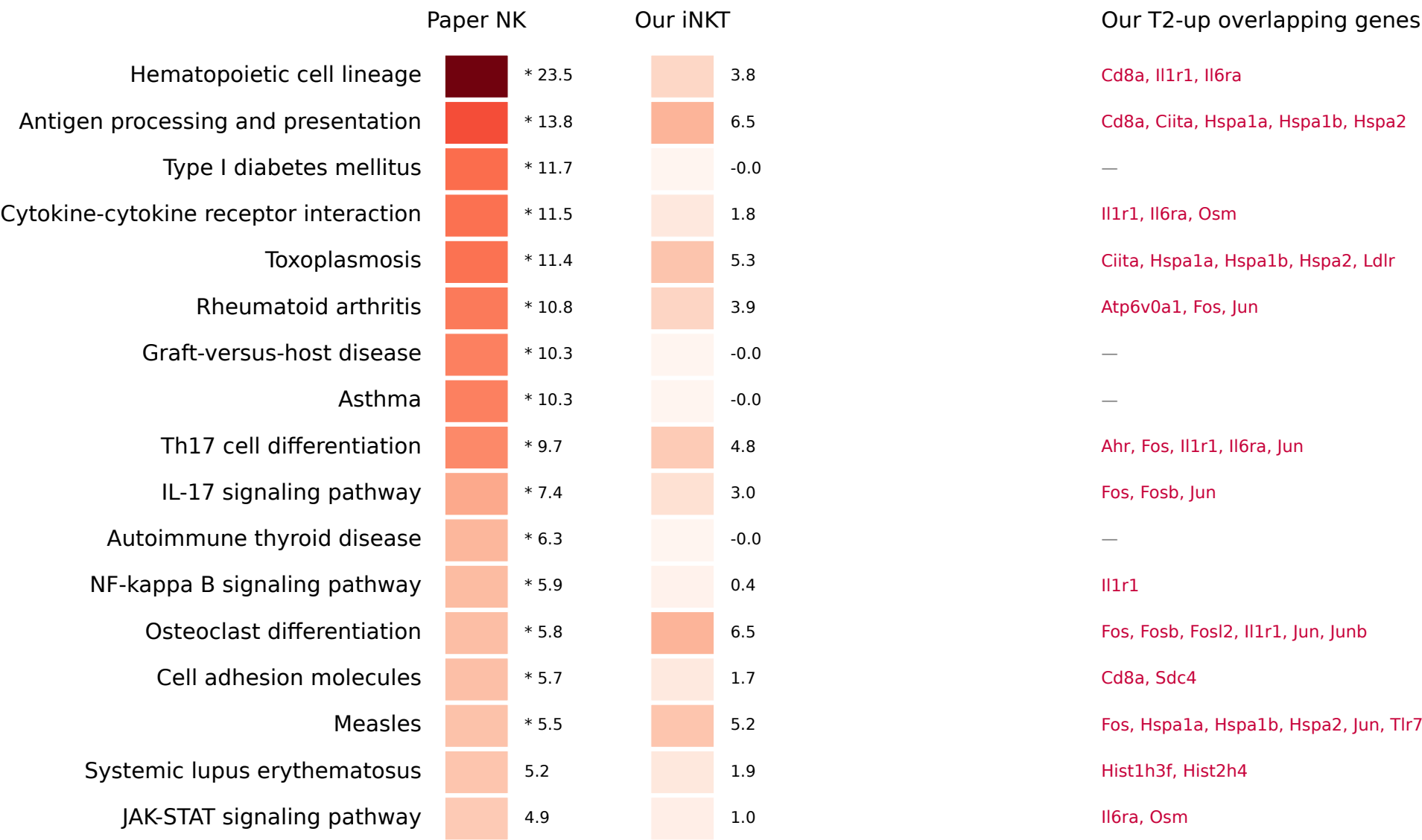
Pathway / reaction	Legacy source	Old FDR	T2-up FDR	Ctrl-up FDR	Current overlapping genes
hsa05020 Prion disease - Homo sapiens (human)	KEGG 2021 HUMAN	0.046	2e-07	1	T2: Cox6c, Hspa1b, Ndufa13 +7 more Ctrl: Hspa8
STIP1(HOP) binds HSP90 and HSP70:HSP40:nascent protein	Reactome 2021	4e-06	0.074	0.0015	T2: Hspa1b Ctrl: Dnaja1, Hspa8
ATP hydrolysis by HSP70	Reactome 2021	0.0018	0.074	0.0015	T2: Hspa1b Ctrl: Dnaja1, Hspa8
HSP70 binds to HSP40:nascent protein	Reactome 2021	0.0018	0.074	0.0015	T2: Hspa1b Ctrl: Dnaja1, Hspa8
HSF1-mediated gene expression	Reactome 2021	0.0027	0.074	0.0015	T2: Hspa1b Ctrl: Hspa8, Hsph1
ATP binding to HSP90 triggers conformation change	Reactome 2021	0.0014	0.074	0.023	T2: Hspa1b Ctrl: Hspa8
FKBP4 binds HSP90:ATP:STIP1:HSP70:nascent protein	Reactome 2021	0.0018	0.074	0.024	T2: Hspa1b Ctrl: Hspa8
FKBP5 binds HSP90:ATP:STIP1:HSP70:nascent protein	Reactome 2021	0.0018	0.074	0.024	T2: Hspa1b Ctrl: Hspa8

Shown: 8 strongest current min(up-FDR, down-FDR), retaining original names; first 3 drivers per direction shown; complete records in exported table.

Legacy/current cluster correspondence is a response comparison, not an identity claim; pathway versions and thresholds differ.

E Paper pathways: C0 / BM

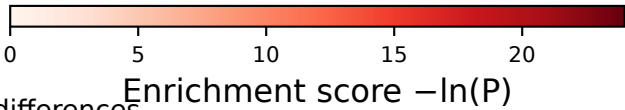
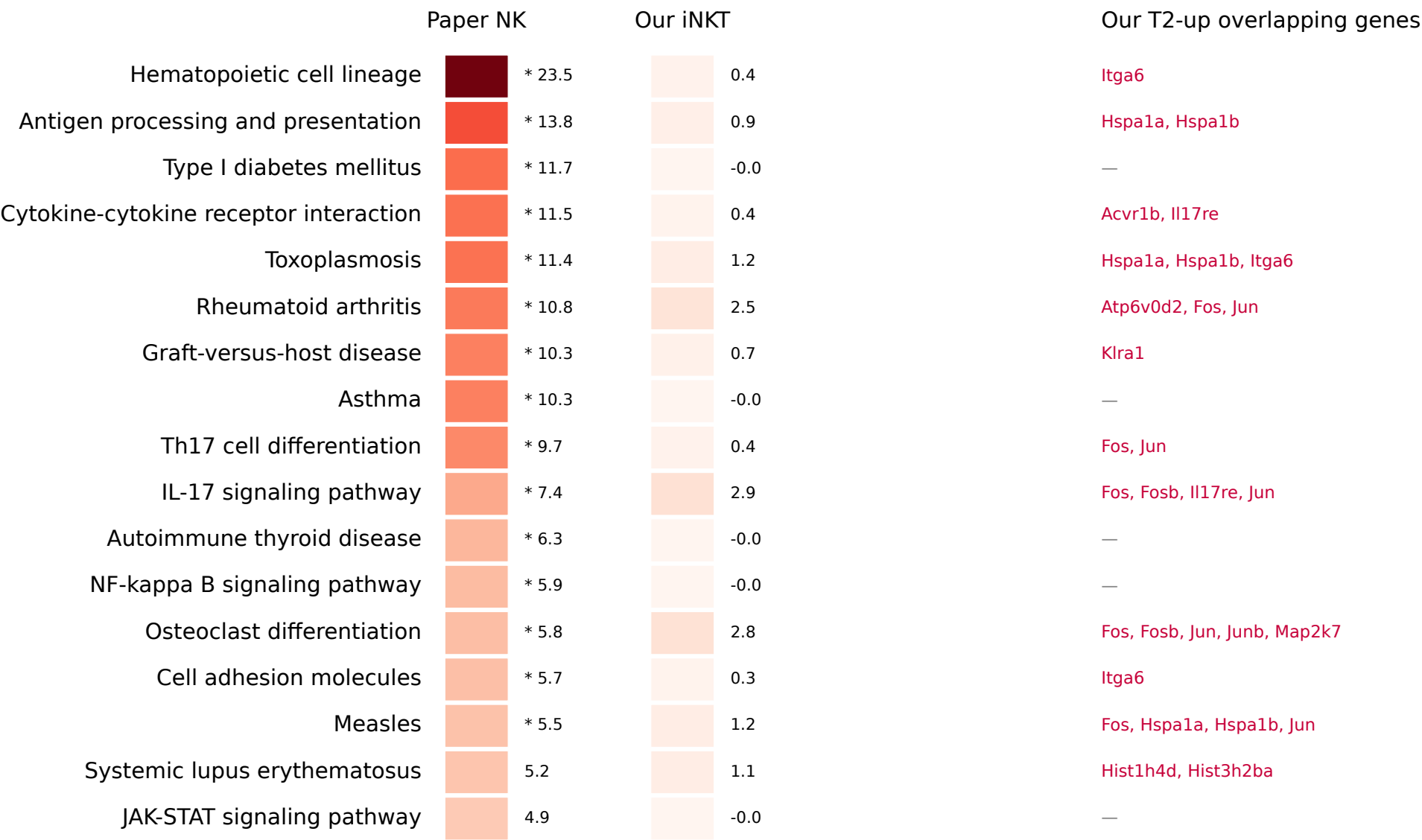
Single enrichment-score columns with actual driver genes; all 17 reference pathways remain visible



Score = $-\ln(\text{nominal enrichment } P)$, verified against Table S3; * pathway BH FDR ≤ 0.05 . Scale does not remove method/library differences.

E Paper pathways: C5-1 / Spl

Single enrichment-score columns with actual driver genes; all 17 reference pathways remain visible



Score = $-\ln(\text{nominal enrichment } P)$, verified against Table S3; * pathway BH FDR ≤ 0.05 . Scale does not remove method/library differences.

E Paper pathways: C6 / Thy

Single enrichment-score columns with actual driver genes; all 17 reference pathways remain visible

	Paper NK		Our iNKT		Our T2-up overlapping genes
Hematopoietic cell lineage	* 23.5		-0.0		—
Antigen processing and presentation	* 13.8		1.4		Hspa1a, Hspa1b
Type I diabetes mellitus	* 11.7		-0.0		—
Cytokine-cytokine receptor interaction	* 11.5		0.2		Ccr4
Toxoplasmosis	* 11.4		1.0		Hspa1a, Hspa1b
Rheumatoid arthritis	* 10.8		-0.0		—
Graft-versus-host disease	* 10.3		-0.0		—
Asthma	* 10.3		-0.0		—
Th17 cell differentiation	* 9.7		-0.0		—
IL-17 signaling pathway	* 7.4		-0.0		—
Autoimmune thyroid disease	* 6.3		-0.0		—
NF-kappa B signaling pathway	* 5.9		-0.0		—
Osteoclast differentiation	* 5.8		-0.0		—
Cell adhesion molecules	* 5.7		0.5		Pecam1
Measles	* 5.5		1.3		Cd46, Hspa1a, Hspa1b
Systemic lupus erythematosus	5.2		-0.0		—
JAK-STAT signaling pathway	4.9		-0.0		—



Score = $-\ln(\text{nominal enrichment } P)$, verified against Table S3; * pathway BH FDR ≤ 0.05 . Scale does not remove method/library differences.

E Current top pathways: C0 / BM

Full-library top 10 T2-up ORA pathways, using the same score-column and gene-callout format

	–ln(P)	T2-up driver genes
MAPK signaling pathway	8.2	Dusp1, Fos, Gadd45g, Hspa1a, Hspa1b, Hspa2, Il1r1, Jun, Nr4a1, Ptpn5
Estrogen signaling pathway	7.1	Fos, Hspa1a, Hspa1b, Hspa2, Jun, Pomc
Antigen processing and presentation	6.5	Cd8a, Ciita, Hspa1a, Hspa1b, Hspa2
Osteoclast differentiation	6.5	Fos, Fosb, Fosl2, Il1r1, Jun, Junb
Toxoplasmosis	5.3	Ciita, Hspa1a, Hspa1b, Hspa2, Ldlr
Measles	5.2	Fos, Hspa1a, Hspa1b, Hspa2, Jun, Tlr7
Aldosterone synthesis and secretion	4.9	Ldlr, Nr4a1, Nr4a2, Pomc
Influenza A	4.9	Ciita, Hspa1a, Hspa1b, Hspa2, Jun, Tlr7
Th17 cell differentiation	4.8	Ahr, Fos, Il1r1, Il6ra, Jun
Fluid shear stress and atherosclerosis	4.7	Dusp1, Fos, Il1r1, Jun, Sdc4

Top 10 by pathway BH FDR, then nominal P and term name; rows without * are not FDR-significant. No post-hoc pathway renaming.

E Current top pathways: C5-1 / Spl

Full-library top 10 T2-up ORA pathways, using the same score-column and gene-callout format

	–ln(P)	T2-up driver genes
Cocaine addiction	6.0	Cdk5r1, Dlg4, Fosb, Jun
Estrogen signaling pathway	4.4	Ebag9, Fos, Hspa1a, Hspa1b, Itpr1, Jun
Fanconi anemia pathway	3.5	Ercc1, Mlh1, Rmi1, Top3b
Mucin type O-glycan biosynthesis	2.9	C1galt1c1, Galnt11
IL-17 signaling pathway	2.9	Fos, Fosb, Il17re, Jun
Folate biosynthesis	2.8	Akr1b10, Mocs1
Mannose type O-glycan biosynthesis	2.8	Pomgnt1, Rxylt1
Osteoclast differentiation	2.8	Fos, Fosb, Jun, Junb, Map2k7
Amphetamine addiction	2.7	Fos, Fosb, Jun
Fluid shear stress and atherosclerosis	2.6	Fos, Gstp1, Jun, Map2k7, Sumo3

Top 10 by pathway BH FDR, then nominal P and term name; rows without * are not FDR-significant. No post-hoc pathway renaming.

E Current top pathways: C6 / Thy

Full-library top 10 T2-up ORA pathways, using the same score-column and gene-callout format

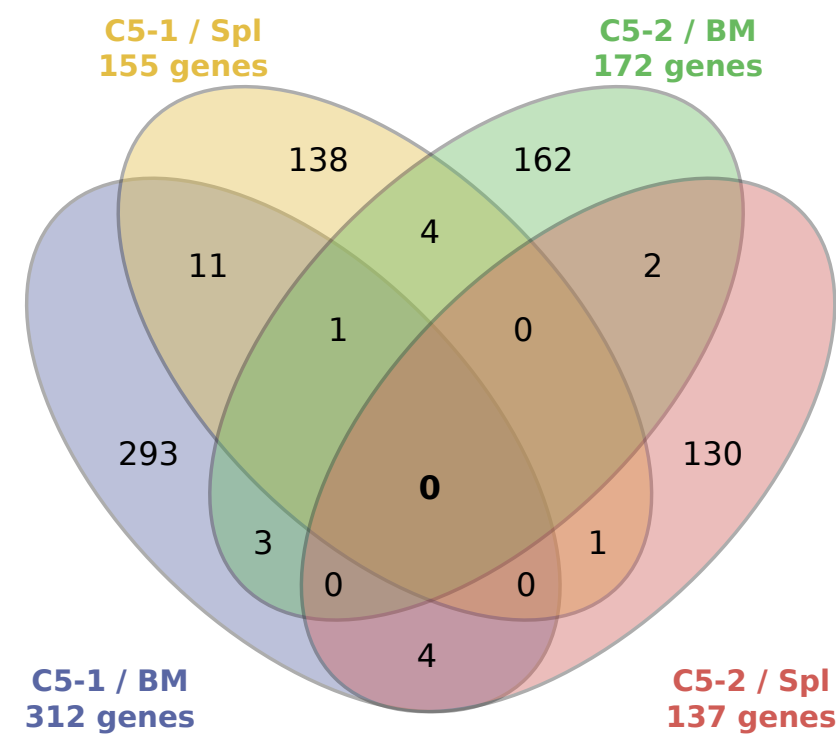
	–ln(P)	T2-up driver genes
Retinol metabolism	4.4	Rdh5, Retsat
Glycine, serine and threonine metabolism	3.1	Chdh, Sardh
Nicotinate and nicotinamide metabolism	3.0	Nmnat2, Nnt
ABC transporters	2.9	Abcb1b, Abcd3
Bile secretion	2.9	Abcb1b, Ephx1
ErbB signaling pathway	2.3	Camk2b, Erbb3, Hbegf
alpha-Linolenic acid metabolism	2.3	Acox1
Peroxisome	2.3	Abcd3, Acox1, Hacl1
PPAR signaling pathway	2.2	Acox1, Sorbs1
Complement and coagulation cascades	2.1	Cd46

Top 10 by pathway BH FDR, then nominal P and term name; rows without * are not FDR-significant. No post-hoc pathway renaming.

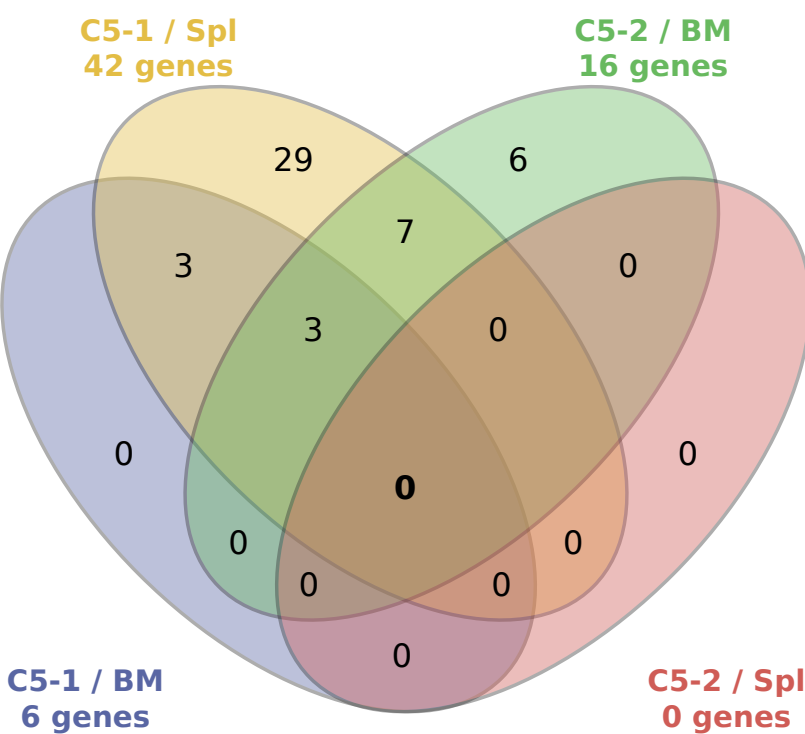
F Shared T2-up genes across state × tissue

Four-set Venn diagrams with exact counts from the existing membership tables

Paper rule: $P < 0.05$, $\log_2FC \geq 1$



Sensitivity: BH FDR ≤ 0.05 , $\log_2FC \geq 0.25$



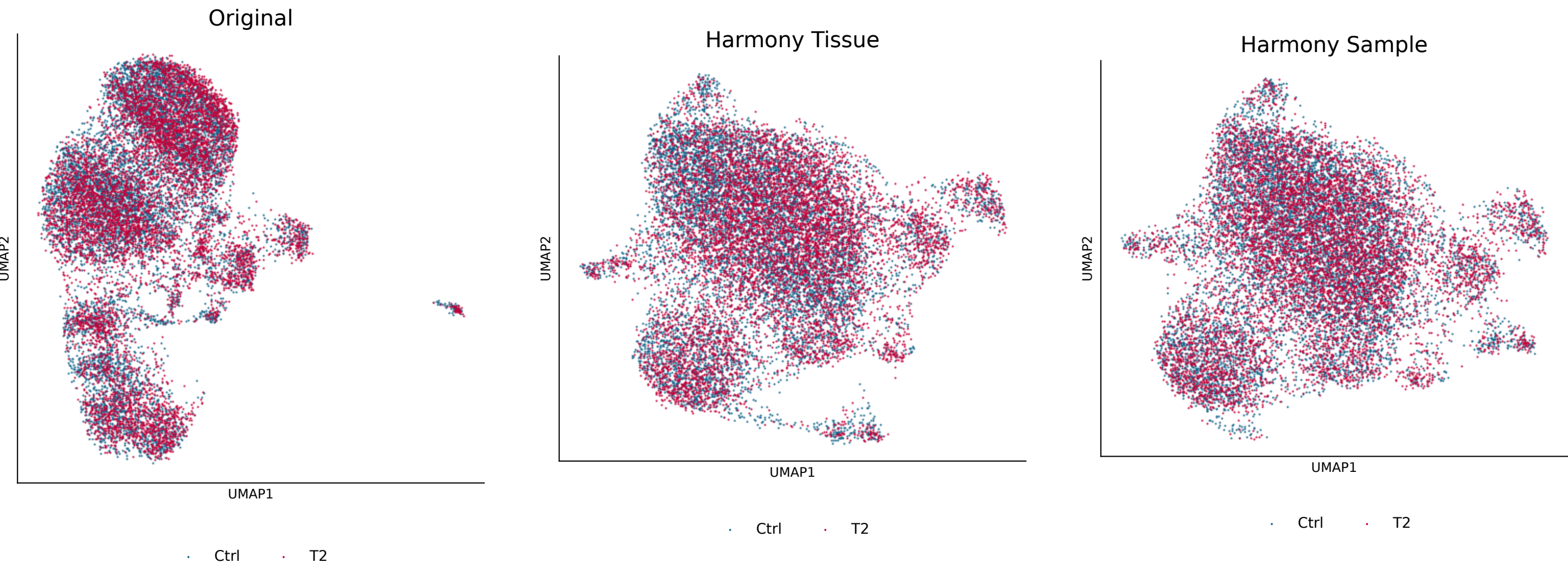
Cell counts (Ctrl/T2): C5-1 BM 124/85; C5-1 Spl 239/345; C5-2 BM 104/96; C5-2 Spl 28/42.

Paper K1/K2 and our C5-1/C5-2 are different populations. Thymus is excluded here because C5-2 Ctrl n=11.

All-four intersection = 0 in both rules. This does not establish absence of shared biology; set membership depends on thresholds.

Batch sensitivity: show the actual UMAPs

Original graph, Harmony by tissue and Harmony by sample

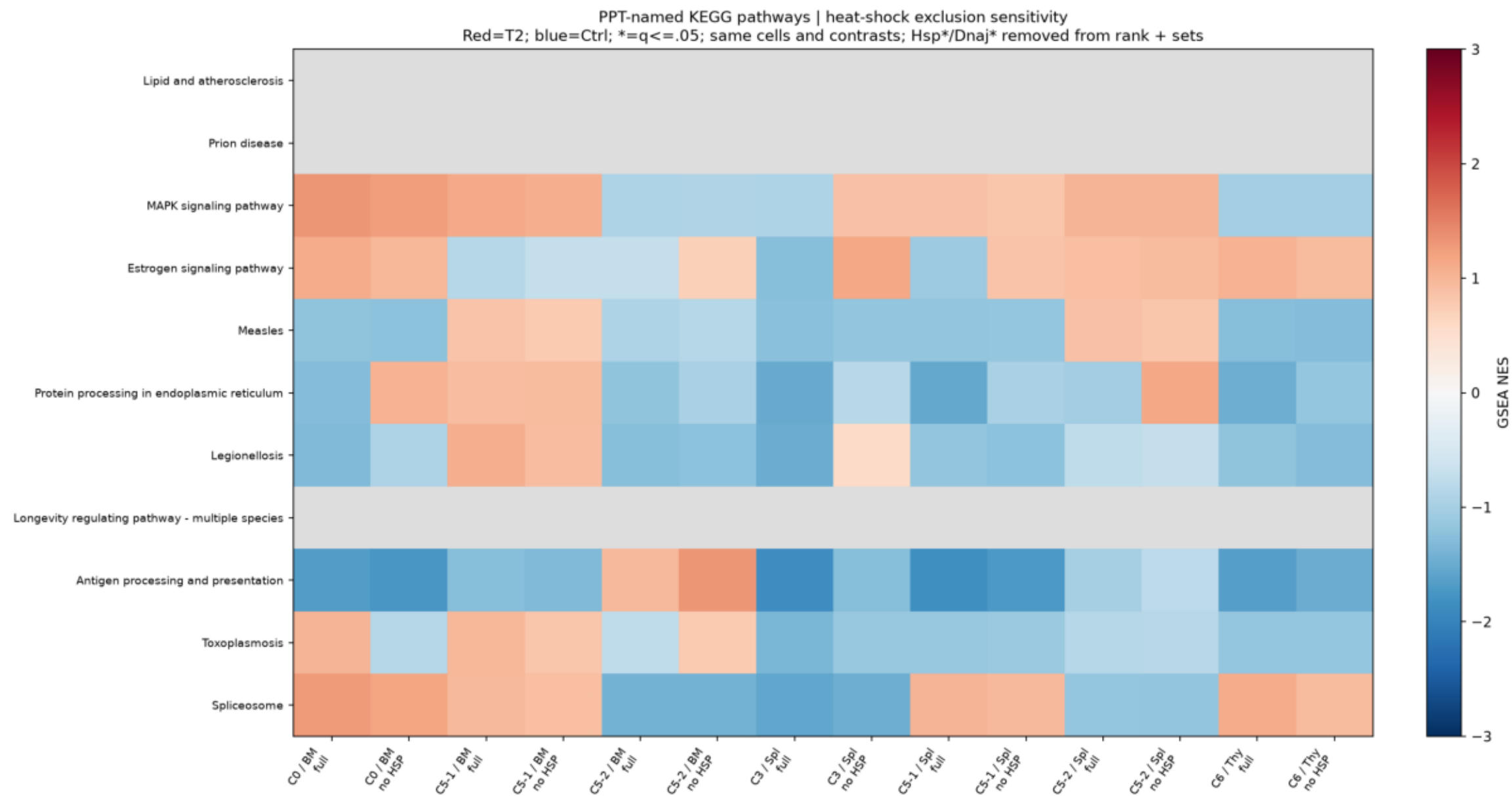


All panels colored by the same condition labels; UMAP coordinates differ between fits. Expression DE still uses uncorrected counts.

Integration increases mixing but changes condition/cluster structure. An independently identified technical batch is unavailable.

Additional analysis: signed GSEA and heat-shock sensitivity

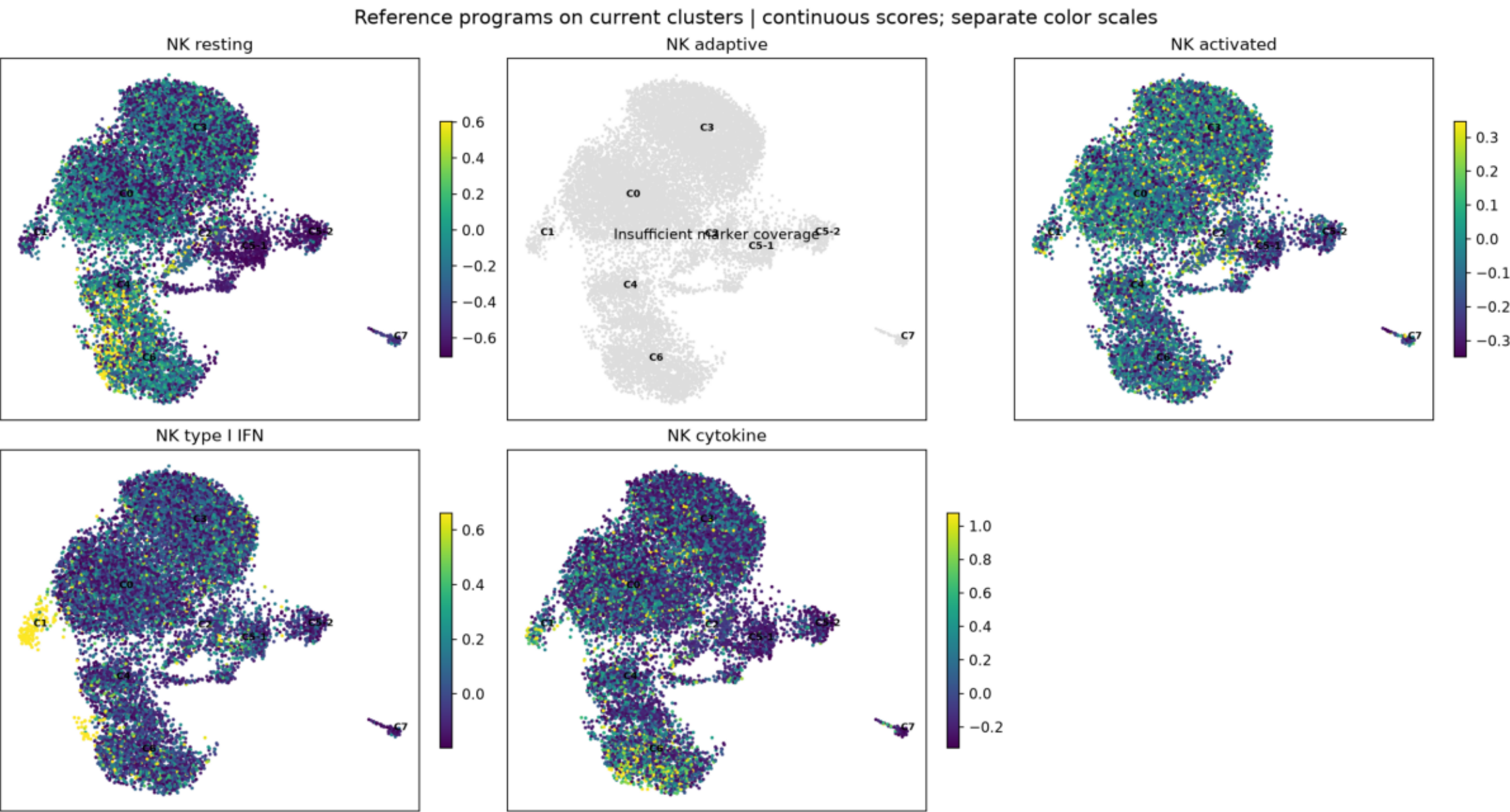
Keep the signed NES heatmap as a separate analysis



Red/blue represent signed NES; symbols use GSEA q. Hsp/Dnaj masking changes the ranking and tested gene sets.

Additional analysis: reference NK state coverage

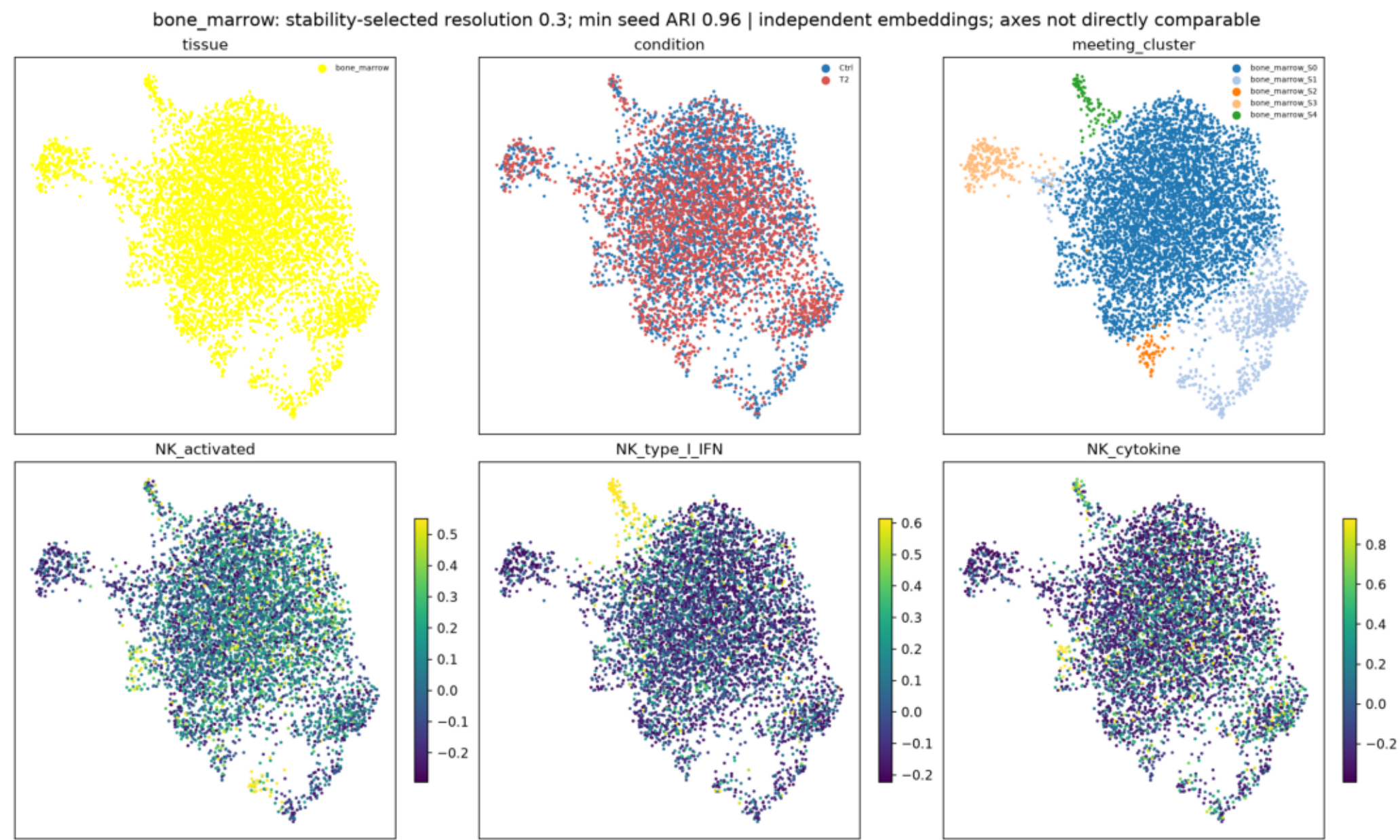
Full-feature sensitivity for the requested state mapping



Adaptive state is not reliably scoreable. These continuous programs do not establish the paper NK cluster identities.

Additional analysis: bone marrow separately

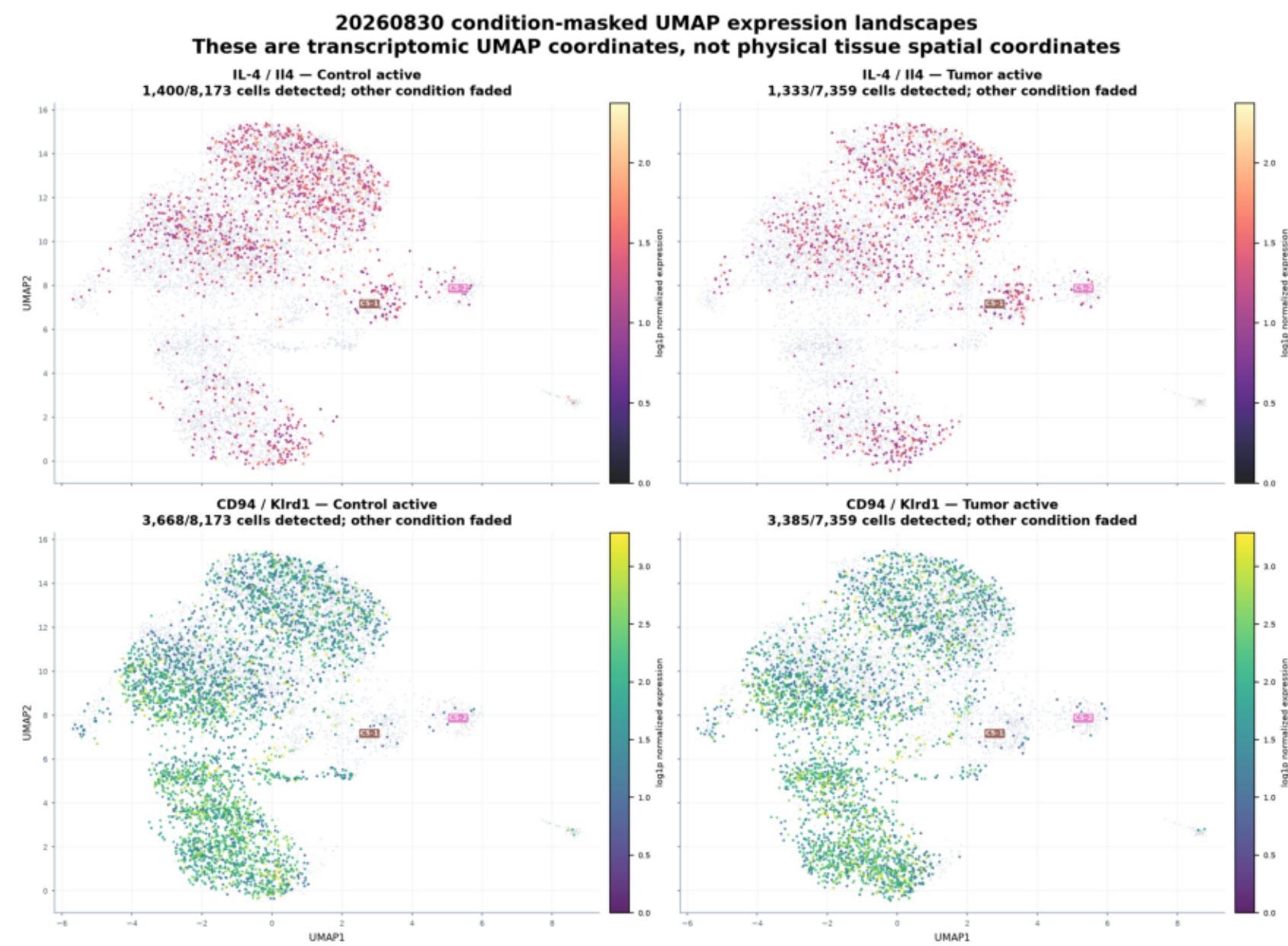
Within-tissue clustering, condition and state maps



New within-tissue labels differ from original C# labels; use saved cell-overlap tables for correspondence.

IL-4 and CD94: condition-masked expression

Both conditions shown using the same saved embedding

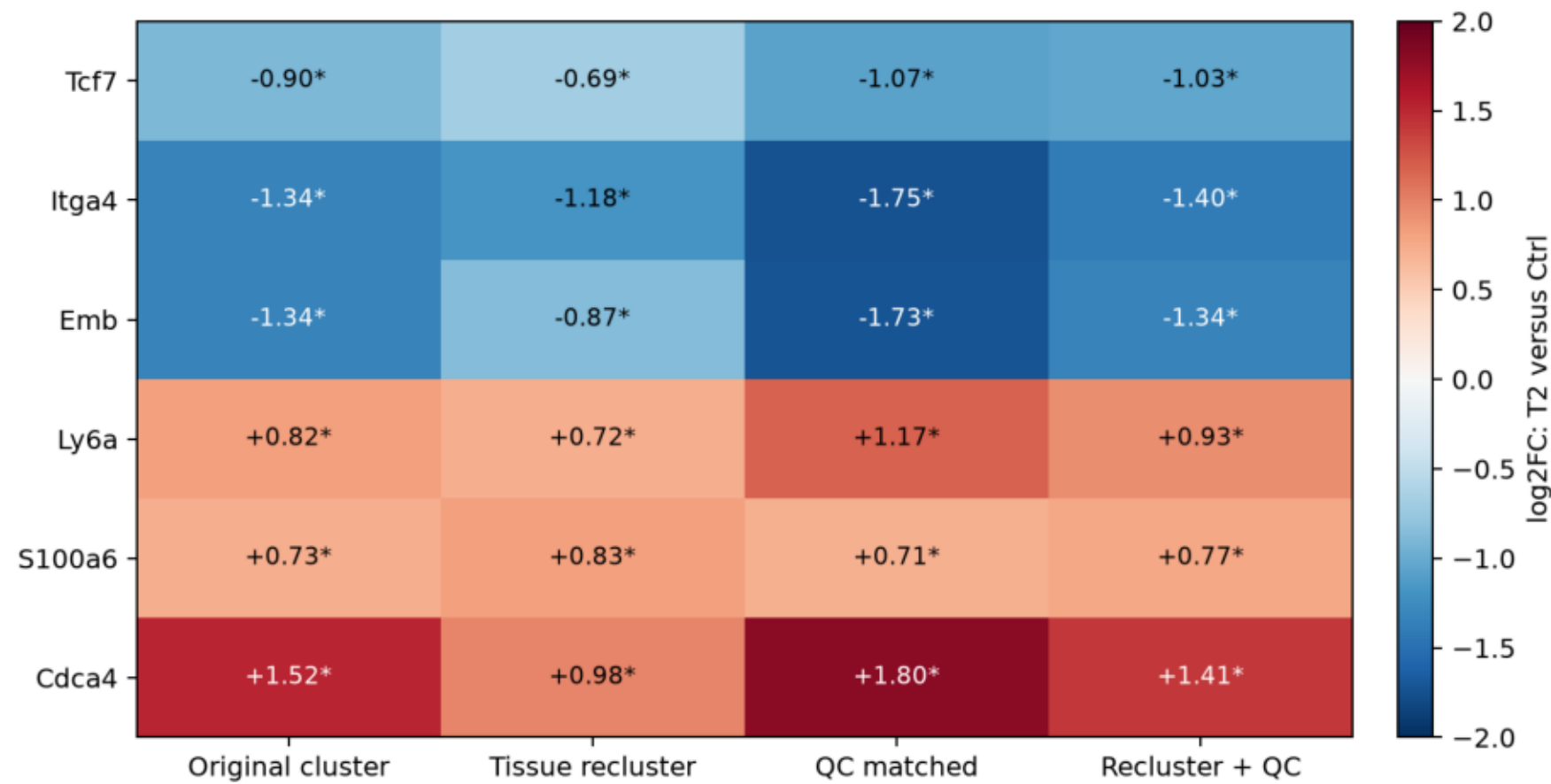


Complementarity is weak, not strict exclusivity. UMAP positions are not physical tissue locations or RNA velocity.

Local information: bone marrow C4

Current candidate expression effects across sensitivity analyses

Priority 1: localized BM C4 expression changes
Ctrl 244 / T2 223 cells; QC matched 164 per condition
98.7% of original C4 maps to BM_S1; BM_S1 also contains other original groups



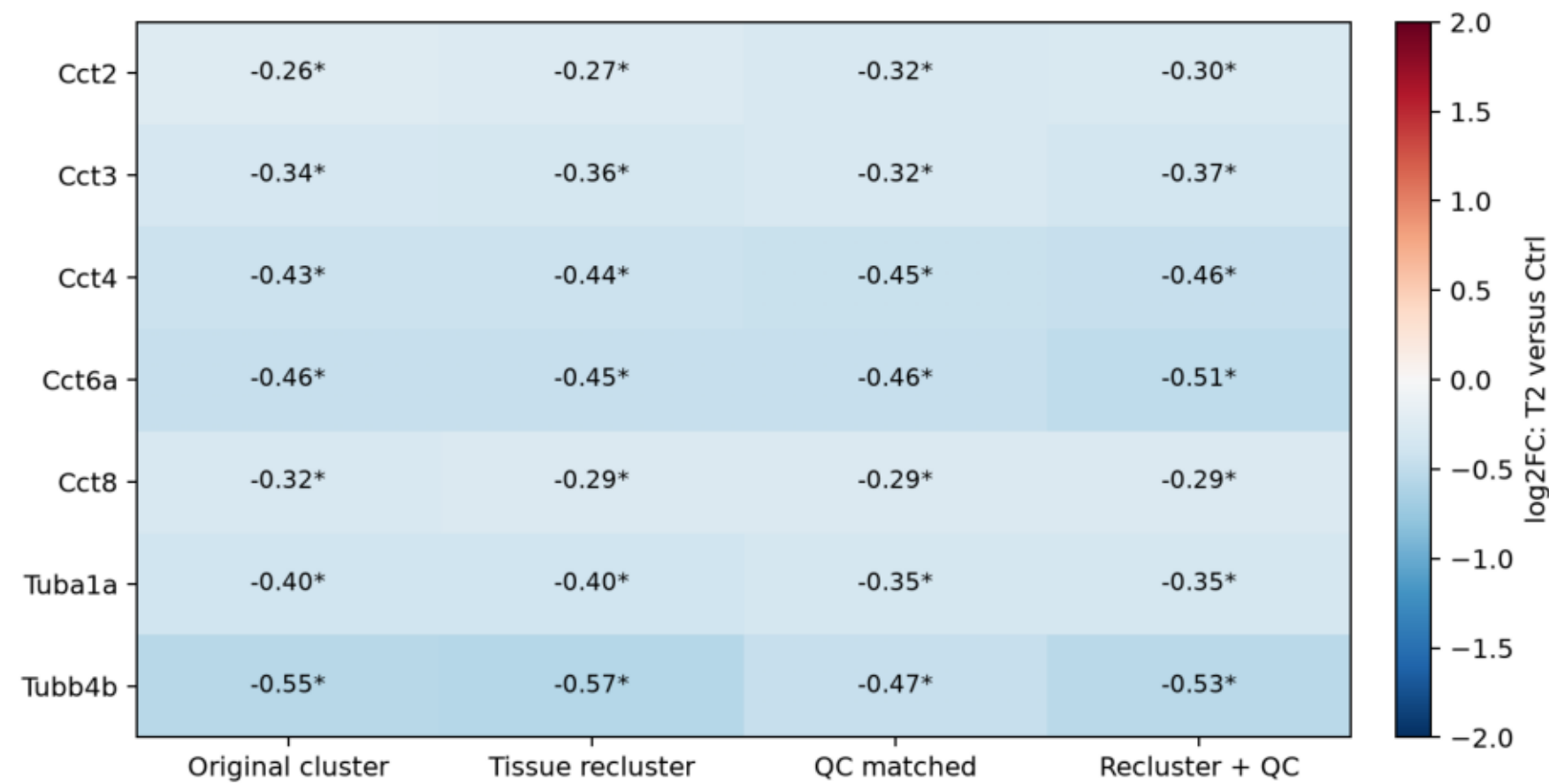
* Gene BH FDR <= 0.05 within each contrast. Columns are sensitivity analyses, not independent replicates.

Same-data sensitivity checks support a candidate, not an independent replicate or established mechanism.

Local information: spleen C3 CCT/TriC program

Specific driver genes and current enrichment evidence

Priority 2: lower CCT/TriC-associated expression in spleen C3
ORA FDR 0.002061; QC 3.976e-05; recluster 0.002744; no Hsp/Dnaj 0.001925
Actin- and tubulin-folding term names share Cct2/3/4/6a/8 drivers



* Gene BH FDR <= 0.05 within each contrast. Columns are sensitivity analyses, not independent replicates.

Shared driver genes link several pathway names. Expression evidence does not directly measure protein-folding or killing function.

What is comparable, and what remains different

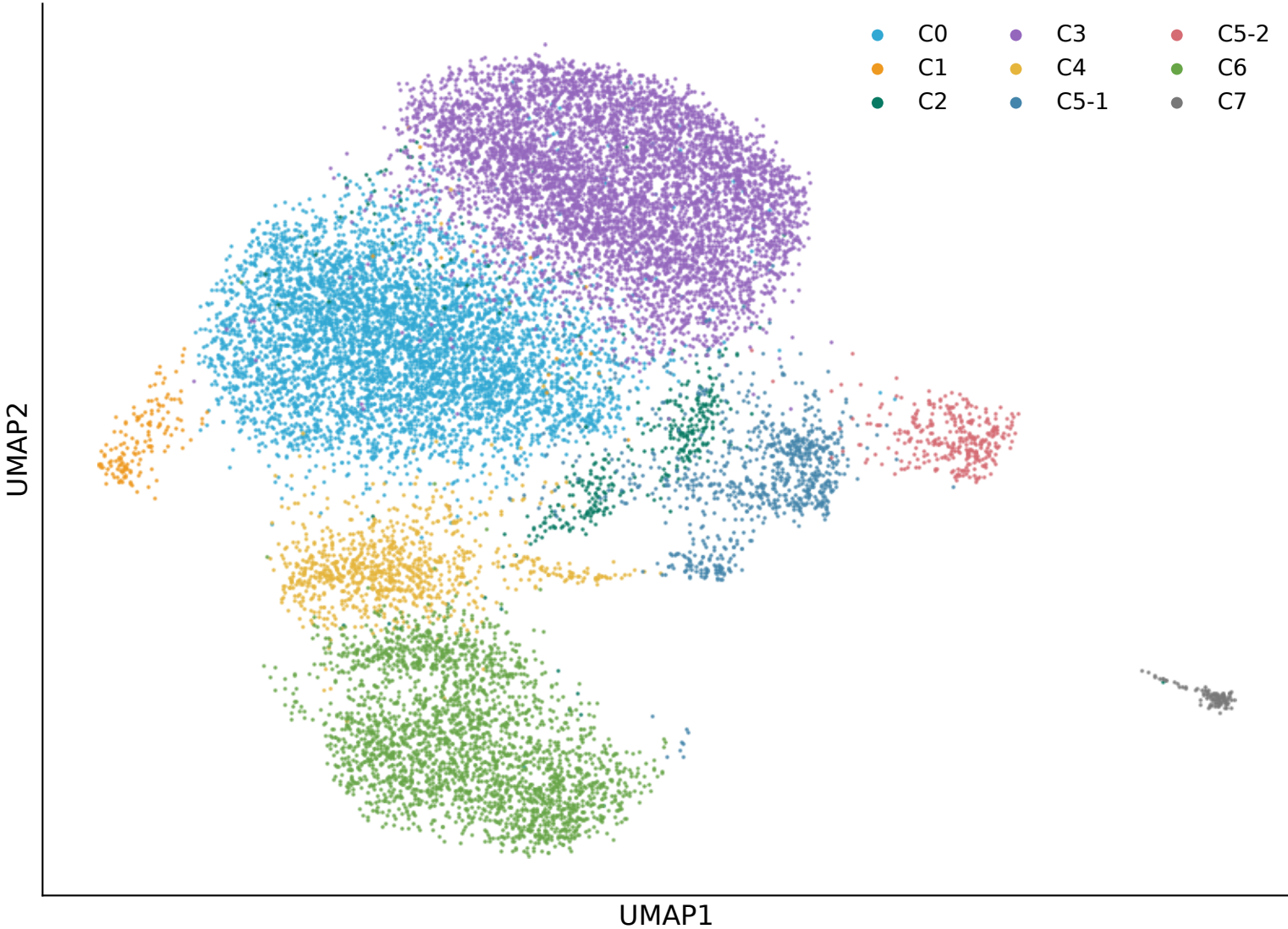
Presentation changes preserve the measured quantity and the population being compared

Category	Interpretation
Matched presentation	Dotplot + DEG table; two composition denominators; volcano axes; ORA score strip; exact four-set Venn.
Different computation	Current UMAP/Leiden vs paper t-SNE/K-means; current module scores vs legacy summed scores.
Different pathway statistics	ORA uses $-\ln(P)$; signed GSEA uses NES. Displaying a term does not imply BH FDR significance.
Different populations	iNKT vs paper NK; C# vs legacy c# or K#. No one-to-one identity assumed.
Not performed / unsupported	RNA velocity, paper qPCR/flow/function assays, PAGER interaction network replication; no replacement figures invented.
Current state of completion	R01-R13 remains as documented: velocity input gap, incomplete state coverage, old-library limitations and no animal-level replication.

Current priority: BM C4 expression-state candidates and Spl C3 CCT/TriC-related expression; independent validation remains pending.

iNKT cytotoxicity and Gene Ontology

Paper-guided analysis on the latest local data | 19 September 2026



15,532 cells | 10,670 measured genes

PAGER: gene modules and temporal profiles

3 tissues | Ctrl and T2 | 9 refined clusters

GAFA: cluster-specific GO enrichment

Primary contrasts: tissue x original cluster

Sensitivity: stable tissue reclustering + QC matching

Exploratory transcriptional results; one sample label per tissue x condition; no animal-level inference.

Evidence summary and next interpretation

Cytotoxic module results, GO support and sensitivity are reported separately.

Contrast	Core delta	BH q (cell level)	QC same direction	Control-score agrees	Max QC SMD
Spl/C1	-0.131	0.059	NA	yes	NA
BM/C2	-0.100	0.22	5/5	yes	0.17
BM/C4	+0.086	0.025	5/5	yes	0.06
Thy/C2	+0.067	0.49	4/5	yes	0.18
Spl/C4	+0.061	0.3	5/5	yes	0.20
Spl/C0	+0.059	0.46	4/5	yes	0.19
Thy/C6	-0.049	0.4	5/5	yes	0.08
Spl/C5-2	-0.035	0.34	NA	yes	NA

Ranked by absolute expression difference, not selected for significance.
A cytotoxic transcript signature is not a killing assay; GO terms can share the same driver genes.
Independent biological replicates and functional assays are required to confirm treatment effects.